

Ch 2: Acids, Bases and Salts

Complete Notes | Deep Definitions | Diagrams | Tables | Exam Tips | Centum Focus

SECTION A - INTRODUCTION, ACIDS, BASES, SALTS AND INDICATORS

1. Introduction to Acids, Bases and Salts

- Acids, bases and salts are important chemical substances that we use in our daily life.
- They have different properties and are used in medicines, foods, cleaning agents, agriculture, industries and laboratories.
- They are also formed naturally in our surroundings and in living organisms.

2. Definition of Acid

- An acid is a substance that turns blue litmus red.
- In water, acids produce hydrogen ions (H^+) which actually exist as hydronium ions (H_3O^+).
- Examples:** HCl (Hydrochloric acid), H_2SO_4 (Sulphuric acid), HNO_3 (Nitric acid), CH_3COOH (Acetic acid).

3. Definition of Base

- A base is a substance that turns red litmus blue.
- In water, bases produce hydroxide ions (OH^-).
- Examples:** NaOH (Sodium hydroxide), KOH (Potassium hydroxide), $Ca(OH)_2$ (Calcium hydroxide), NH_4OH (Aqueous ammonia).

4. Definition of Salt

- A salt is a substance formed when an acid reacts with a base. The reaction is called neutralisation.
- Salts are usually ionic compounds.
- Examples:** NaCl (Sodium chloride), Na_2CO_3 (Sodium carbonate), $CaCO_3$ (Calcium carbonate).

5. Natural/Common Examples in Daily Life

Acids		Bases	
Substance	Source / Use	Substance	Source / Use
Citric acid	Lemon, orange, citrus fruits	Sodium bicarbonate	Baking soda
Lactic acid	Curd, sour milk	Calcium hydroxide	Slaked lime, used in whitewashing
Acetic acid	Vinegar	Magnesium hydroxide	Milk of magnesia (antacid)
Tartaric acid	Tamarind, grapes	Sodium hydroxide	Soap making, drain cleaner
Formic acid	Ant sting, nettle leaves	Aqueous ammonia	Window cleaner

6. Indicators – Meaning, Need and Types

- Indicators** are substances that show different colours in acidic and basic solutions.
- Need:** Many acids and bases are colourless. Indicators help us identify whether a solution is acidic, basic or neutral.
- Types of Indicators:**
 - Natural indicators** – Litmus, turmeric.
 - Synthetic indicators** – Phenolphthalein, Methyl orange.
 - Olfactory indicators** – Onion, vanilla essence (These indicators change smell in acidic/basic medium).

7. Indicator Colour Changes

Indicator	In Acidic Solution	In Basic Solution	In Neutral Solution
Litmus	Red	Blue	Purple
Turmeric (natural)	Yellow	Reddish brown	Yellow
Phenolphthalein	Colourless	Pink	Colourless
Methyl orange	Red	Yellow	Orange
Olfactory indicators (Onion / Vanilla)	No smell	Pungent smell	Faint or no change

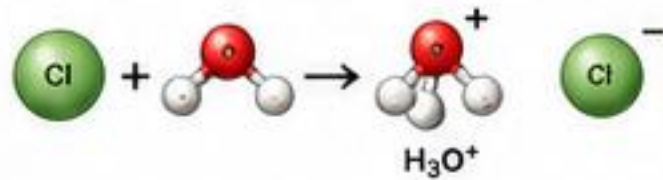
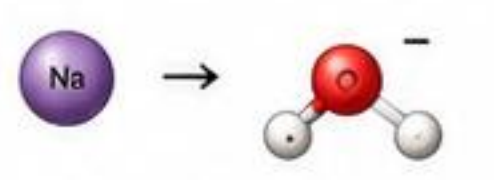
8. Properties of Acids

- Sour taste (e.g., lemon juice).
- Turn blue litmus red.
- In water, they produce H^+ (as H_3O^+) ions.
- Their aqueous solutions conduct electricity.
- React with metals to produce salt and hydrogen gas.
- React with bases (alkalis) to form salt and water (neutralisation).
- React with carbonates / bicarbonates to produce salt, water and CO_2 .
- Generally corrosive in nature.

9. Properties of Bases

- Bitter taste (e.g., baking soda solution).
- Soapy / slippery feel.
- Turn red litmus blue.
- In water, they produce OH^- ions.
- Their aqueous solutions conduct electricity.
- React with acids to form salt and water (neutralisation).
- Generally caustic / corrosive.

10. Acid and Base in Water – Ion Formation

Acid in Water (e.g., HCl)		Base in Water (e.g., NaOH)	
$HCl + H_2O \rightarrow H_3O^+ + Cl^-$		$NaOH \rightarrow Na^+ + OH^-$	
			
HCl (Acid)	H ₂ O (Water)	H ₃ O ⁺ (Hydronium ion)	Cl ⁻ (Chloride ion)
		Na ⁺ (Sodium ion)	OH ⁻ (Hydroxide ion)

11. Quick Comparison

Feature	Acid	Base	Salt
Definition	Turns blue litmus red; gives H^+ / H_3O^+ in water	Turns red litmus blue; gives OH^- in water	Formed by reaction between an acid and a base
Taste	Sour	Bitter	Generally salty
Litmus Effect	Blue $\rightarrow$ Red	Red $\rightarrow$ Blue	No change (Neutral)
Ions in Water	H^+ / H_3O^+	OH^-	Depends on salt
Conductivity	Conducts electricity	Conducts electricity	Conducts (ionic solutions)
Example	HCl, H_2SO_4	NaOH, $Ca(OH)_2$	NaCl, Na_2CO_3

EXAM TIPS – CENTUM FOCUS

- ★ Learn definitions word by word. Write key ions: H^+ , H_3O^+ (acid) and OH^- (base).
- 🔑 Memorise indicator colour changes in a table.
- ✍ Write examples from daily life to score better in long answers.
- ✅ Learn properties with reactions (acid + base, acid + metal, acid + carbonate).
- 🧠 Practice short answers and diagrams like ion formation for full marks.

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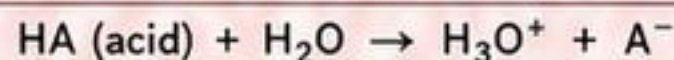
SECTION B - CHEMICAL REACTIONS OF ACIDS AND BASES

1. Why Do Acids and Bases React the Way They Do?

- Acids contain H^+ ions and bases contain OH^- ions.
- When they react, H^+ and OH^- ions combine to form water.
- Other ions present form salts.
- This is why many reactions of acids and bases produce salt and water.

2. Acids and Bases in Water

- Acids in water produce hydronium ions (H_3O^+).



- Bases in water produce hydroxide ions (OH^-).



- These ions are responsible for the acidic or basic nature of solutions.

3. Why Dry HCl Does Not Change Dry Blue Litmus

- Dry HCl gas does not ionise in the absence of water.
- Without water, no H_3O^+ ions are formed.
- Therefore, dry HCl does not change dry blue litmus.

4. Reaction of Acids with Metals



Examples (Balanced Equations):

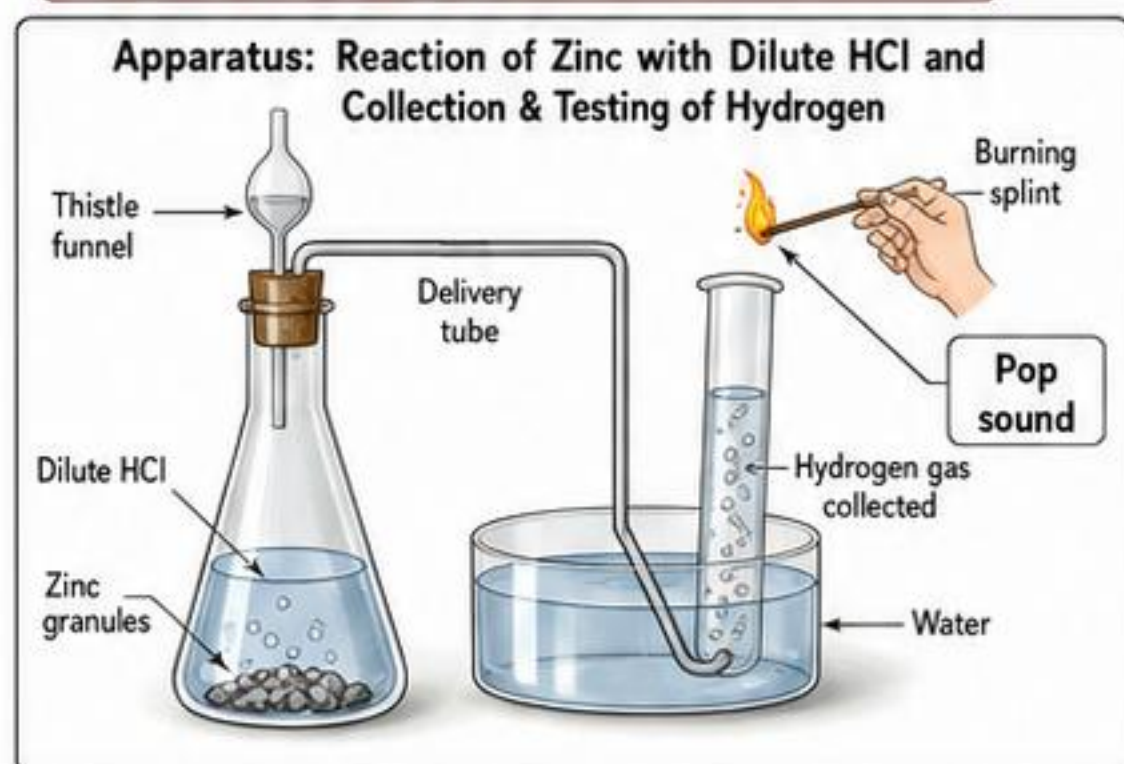
- $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
- $Mg + H_2SO_4 \rightarrow MgSO_4 + H_2$
- $Fe + 2HCl \rightarrow FeCl_2 + H_2$

Observations

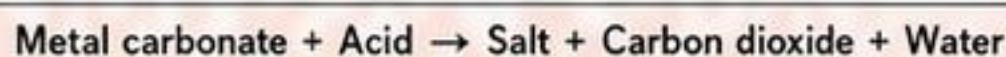
- Effervescence (bubbles) is seen.
- The metal gradually dissolves.
- The test for hydrogen gas is positive (pop sound).

Pop Sound Test for Hydrogen

Bring a burning splint near the gas collected. It burns with a pop sound.



5. Reaction of Metal Carbonates with Acids



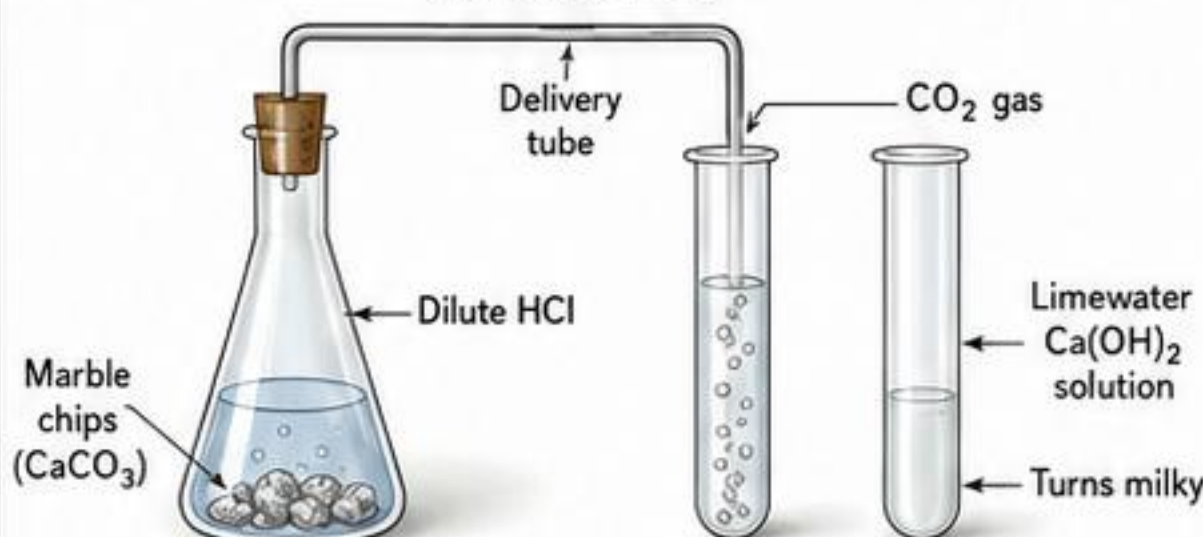
Examples (Balanced Equations):

- $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$
- $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$

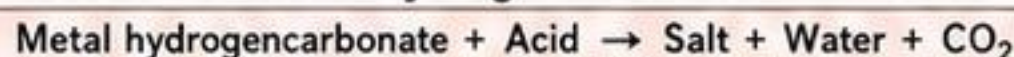
Test for Carbon Dioxide

- Bubble the gas through limewater ($Ca(OH)_2$ solution).
- Limewater turns milky due to formation of $CaCO_3$.
- If CO_2 is passed in excess, the milky disappears (formation of soluble calcium hydrogen carbonate).

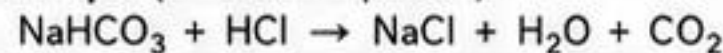
Apparatus: Reaction of Marble ($CaCO_3$) with Dilute HCl and Test for CO_2



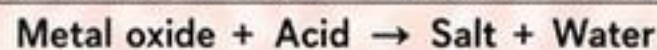
6. Reaction of Metal Hydrogencarbonates with Acids



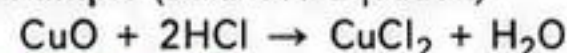
Example (Balanced Equation):



7. Reaction of Metal Oxides with Acids

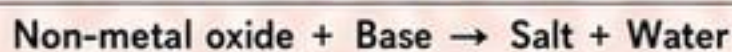


Example (Balanced Equation):



Observation: The black solid CuO dissolves and a blue-green solution of copper(II) chloride is formed.

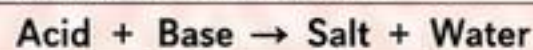
8. Reaction of Non-Metal Oxides with Bases



Examples (Balanced Equations):

- $CO_2 + Ca(OH)_2 \rightarrow CaCO_3 + H_2O$
- $CO_2 + 2NaOH \rightarrow Na_2CO_3 + H_2O$

9. Neutralization Reaction



Examples (Balanced Equations):

- $HCl + NaOH \rightarrow NaCl + H_2O$
- $H_2SO_4 + 2KOH \rightarrow K_2SO_4 + 2H_2O$

Everyday Uses of Neutralization

- Antacids neutralize excess acid in the stomach.
- Acidic soil is treated with bases like slaked lime ($Ca(OH)_2$).
- Bee stings (acidic) are relieved by baking soda (basic).

10. Summary – How Acids React with Different Substances

Substance	General Reaction	Example (Balanced)	Key Observation
Metals	$\text{Acid} + \text{Metal} \rightarrow \text{Salt} + H_2$	$Zn + 2HCl \rightarrow ZnCl_2 + H_2$	Bubbles of H_2 ; metal dissolves
Metal carbonates	$\text{Acid} + \text{Carbonate} \rightarrow \text{Salt} + CO_2 + H_2O$	$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$	Effervescence; CO_2 turns limewater milky
Metal hydrogencarbonates	$\text{Acid} + \text{Hydrogencarbonate} \rightarrow \text{Salt} + H_2O + CO_2$	$NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2$	Effervescence due to CO_2
Metal oxides	$\text{Acid} + \text{Metal oxide} \rightarrow \text{Salt} + H_2O$	$CuO + 2HCl \rightarrow CuCl_2 + H_2O$	Solid dissolves; solution forms
Bases	$\text{Acid} + \text{Base} \rightarrow \text{Salt} + H_2O$	$HCl + NaOH \rightarrow NaCl + H_2O$	Neutralization; heat may evolve
Non-metal oxides	$\text{Non-metal oxide} + \text{Base} \rightarrow \text{Salt} + H_2O$	$CO_2 + Ca(OH)_2 \rightarrow CaCO_3 + H_2O$	Precipitate forms (e.g., $CaCO_3$ white)

EXAM TIPS – CENTUM FOCUS

- Learn key reactions with balanced equations and conditions.
- Always write observations and tests (pop sound for H_2 , limewater test for CO_2).
- Link reactions to real-life uses – helps in long answers.
- Practice diagram-based questions – apparatus and gas collection are important in exams.

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SECTION C - STRENGTH OF ACIDS AND BASES, pH SCALE AND ITS IMPORTANCE

1. Concentrated vs Dilute Solutions

Concentrated Solution	Dilute Solution
Large amount of solute	Small amount of solute
Small amount of solvent	Large amount of solvent
More particles per unit volume	Less particles per unit volume
More acidic/basic in nature*	Less acidic/basic in nature*

* Strength is not decided by concentration. See next section.

2. Strength of Acids and Bases

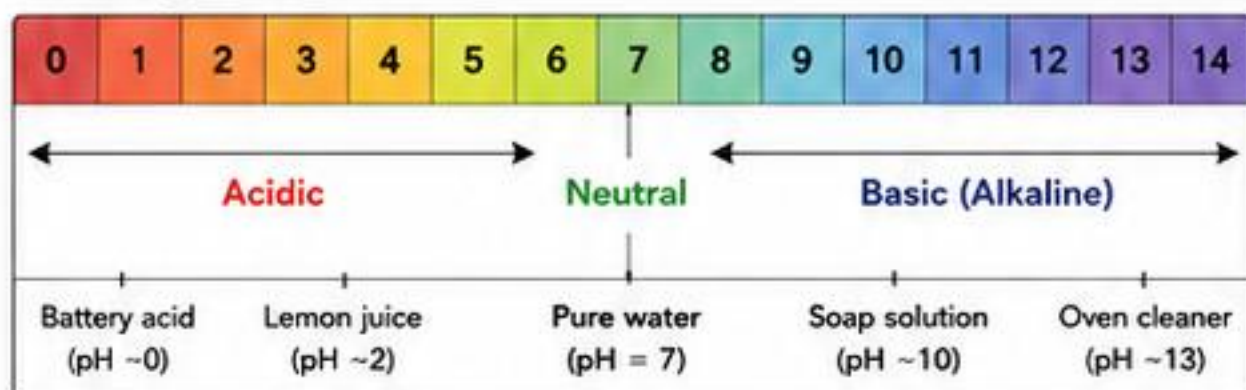
- The strength of an acid or base depends on the **degree of ionization** in water, not on its concentration.
- Strong acids/bases ionize almost completely in water. Weak acids/bases ionize only partially.

Acids		Bases	
Strong Acids (ionize completely)	Weak Acids (ionize partially)	Strong Bases (ionize almost completely)	Weak Bases (ionize partially)
• HCl (Hydrochloric acid)	• CH ₃ COOH (Acetic acid)	• NaOH (Sodium hydroxide)	• NH ₄ OH (Ammonium hydroxide)
• HNO ₃ (Nitric acid)	• H ₂ CO ₃ (Carbonic acid)	• KOH (Potassium hydroxide)	• Mg(OH) ₂ (Magnesium hydroxide)
• H ₂ SO ₄ (Sulphuric acid)		• Ca(OH) ₂ (Calcium hydroxide)	

3. pH and Universal Indicator

- Universal indicator is a mixture of indicators which shows a series of colours with change in pH.
- pH (potential of hydrogen) expresses the concentration of H⁺ ions in a solution. pH scale ranges from 0 to 14.

4. The pH Scale



5. Common pH Values

Substance	Approx. pH	Nature	Indicator Colour (Universal Indicator)
Stomach acid	1 – 2	Acidic	Red
Lemon juice	2	Acidic	Orange
Milk	6 – 6.5	Slightly acidic	Yellowish green
Pure water	7	Neutral	Green
Blood	7.4	Slightly basic	Bluish green
Soap solution	10 – 11	Basic	Blue
Baking soda solution	8.3	Slightly basic	Blue-green

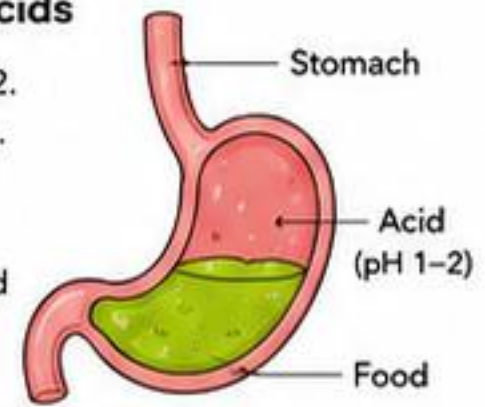
10. Safety Rule – Dilution of Acids

- Always add acid to water, never add water to acid.

Correct Method (Add acid to water)	Wrong Method (Add water to acid)
✓ Heat produced is controlled. ✓ No splashing. ✓ Safe method.	✗ Heat produced suddenly. ✗ Acid splashes out. ✗ Dangerous.

6. Importance of pH – Digestion and Antacids

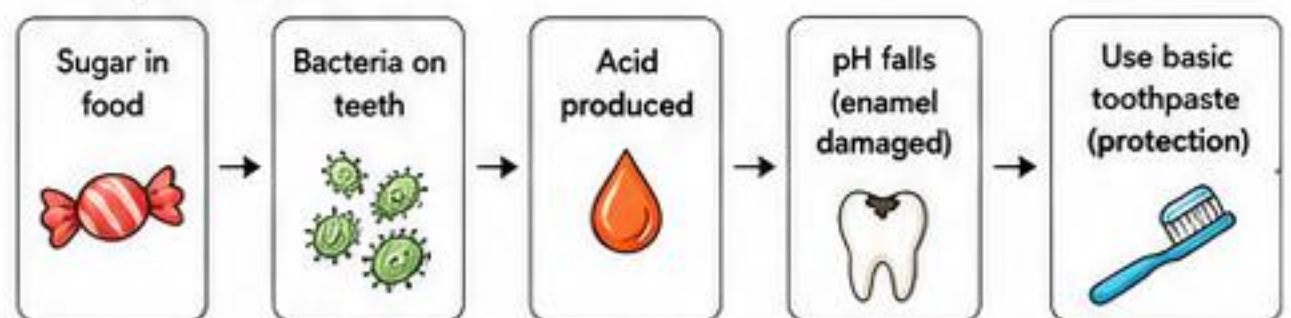
- Hydrochloric acid (HCl) in the stomach has pH 1 – 2. It helps in digestion of food and kills harmful germs.
- Excess acid causes acidity and heartburn.
- Antacids are mild bases which neutralize excess acid and give relief.



Example of antacids: Milk of magnesia (Mg(OH)₂), Baking soda (NaHCO₃), Aluminium hydroxide [Al(OH)₃].

7. pH Change in Tooth Decay

- Sugars in food remain on teeth.
- Bacteria convert sugar into acid.
- Acid lowers pH and damages tooth enamel.
- Brushing with basic toothpaste neutralizes acid and protects teeth.



8. Self-Defence by Animals and Plants

Living Organism	Chemical Used (Self-defence)	Effect	Relief / Treatment
Ant (Red ant)	Formic acid	Pain, irritation	Apply baking soda solution
Honey bee	Formic acid	Pain, swelling	Apply baking soda solution
Wasp	Methanoic acid (Formic acid)	Severe pain, burning	Apply baking soda solution
Nettle plant leaves	Methanoic acid in tiny hairs	Itching, burning	Rub with soap solution

9. Soil pH and Plant Growth

- Different plants grow best in different soil pH.
- Most plants grow well in slightly acidic to neutral soil (pH 6.0 – 7.5).
- Acidic soil (pH < 5.5) can be treated with agricultural lime (CaCO₃). Organic matter like compost or manure also improves soil.

Soil pH	Nature	Effect on Plants	Treatment
Below 5.5	Acidic	Poor growth	Add lime (CaCO ₃)
6.0 – 7.5	Slightly acidic to neutral	Best growth	No treatment needed
Above 8.0	Alkaline	Poor growth	Add organic matter

11. Summary – Quick Comparison

Topic	Key Point	Examples / Range
Concentration	Amount of solute in a given amount of solution.	Concentrated vs Dilute
Strength	Depends on degree of ionization, not concentration.	HCl strong, CH ₃ COOH weak
pH Scale	0 (strong acid) to 14 (strong base); 7 is neutral.	See pH scale
Importance	Affects digestion, tooth health, soil fertility, life processes.	Stomach pH, tooth decay, plant growth
Safety	Always add acid to water while diluting.	Add acid → water (safe)

EXAM TIPS – CENTUM FOCUS

- ★ Learn and remember the pH scale colours and examples.
- ✏ Draw and label the pH scale neatly in exams.
- ✓ Give reasons: strength depends on ionization, not concentration.
- 🧠 Link concepts to daily life: digestion, tooth decay, soil, self-defence.

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SECTION D - SALTS, COMMON SALT AND THE CHLOR-ALKALI PROCESS

1. What is a Salt? How are Salts Formed?

- A salt is an ionic compound formed when the H^+ ion(s) of an acid are completely or partially replaced by a metal or NH_4^+ ion.
- Most salts are formed by the reaction (neutralization) between an acid and a base.



2. Classification of Salts

- Neutral Salts:** Formed when all H^+ of the acid is replaced.
Examples: $NaCl$ (from $HCl + NaOH$), Na_2SO_4 (from H_2SO_4)
- Acidic Salts:** Formed when only some H^+ of a polybasic acid is replaced.
Examples: $NaHSO_4$ (from H_2SO_4), $NaHCO_3$ (from H_2CO_3)
- Basic Salts:** Formed when OH^- ions remain in the salt.
Examples: $Mg(OH)Cl$ (from $Mg(OH)_2 + HCl$),
 $Pb(OH)Cl$ (from $Pb(OH)_2 + HCl$)

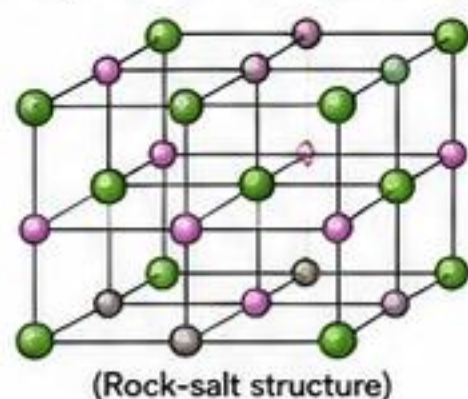
3. Families of Salts

Family of Salts	General Formula	Parent Acid	Parent Base
Chlorides	MX, MCl_n	HCl (Hydrochloric acid)	MOH
Sulphates	M_2SO_4 or MSO_4	H_2SO_4 (Sulphuric acid)	$M(OH)_2$
Nitrates	MNO_3	HNO_3 (Nitric acid)	MOH
Carbonates	M_2CO_3 or MCO_3	H_2CO_3 (Carbonic acid)	$M(OH)_2$
Hydrogencarbonates (Bicarbonates)	$MHCO_3$	H_2CO_3 (Carbonic acid)	MOH

4. General Properties of Salts

- Salts are ionic compounds.
- They are crystalline solids at room temperature.
- They have high melting and boiling points.
- Most salts are soluble in water.
- Solid salts do not conduct electricity, but their molten state and aqueous solutions conduct electricity due to free ions.

5. Crystal Lattice of Sodium Chloride ($NaCl$)

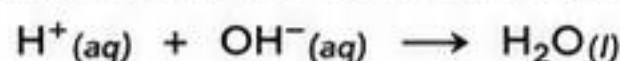


Each Na^+ ion is surrounded by 6 Cl^- ions and each Cl^- ion is surrounded by 6 Na^+ ions.

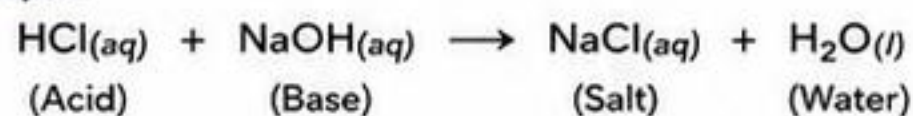
(Rock-salt structure)

6. Neutralization – A Method of Salt Formation

- When an acid reacts with a base, the H^+ of the acid and OH^- of the base combine to form water and the remaining ions form salt.



- Example:



7. Common Salt – Sodium Chloride ($NaCl$)

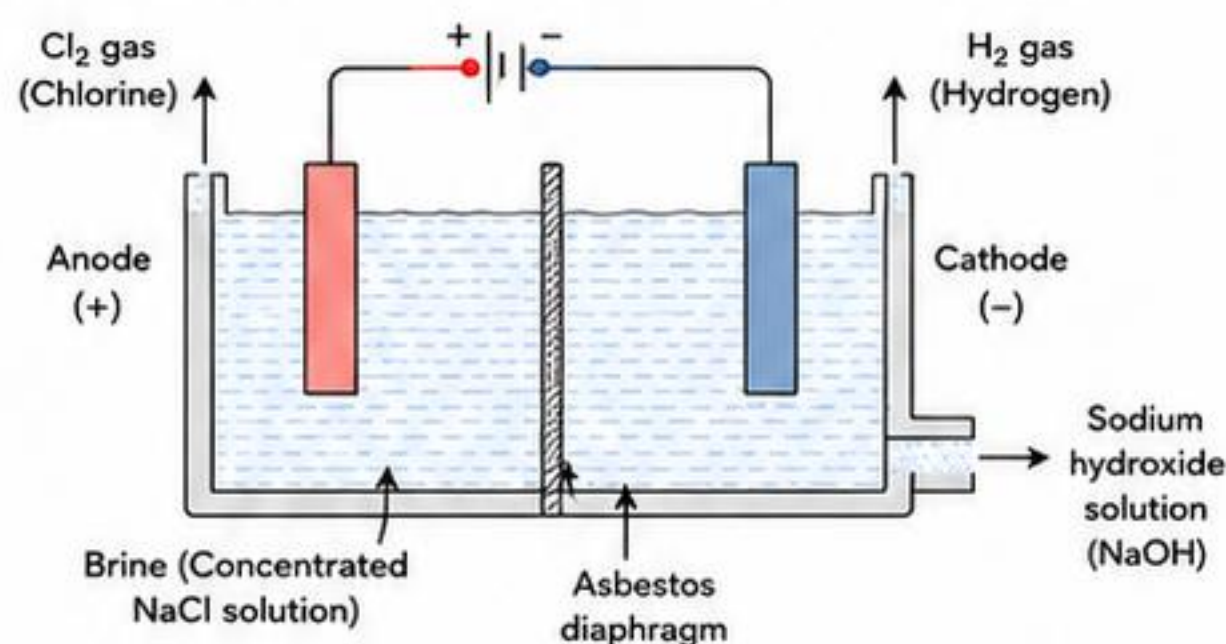
- Sources:** Occurs in nature in two main forms.
 - Sea water:** Contains about 2.5% salt.
 - Rock salt (Halite):** Mined from salt deposits in the earth.
- Preparation (from sea water):**
 - Sea water is collected in shallow beds (salt pans).
 - Water evaporates in the sun.
 - Crystals of common salt remain and are collected.
- Importance and Uses:**
 - Essential for life. Regulates water balance in our body.
 - Used as a preservative in food.
 - Used in curing hides and skins.
 - Raw material in many chemical industries.
 - Used in preparing sodium hydroxide, chlorine and hydrogen.

8. The Chlor-Alkali Process

- It is the electrolysis of concentrated solution of sodium chloride (brine) using a diaphragm cell.
- Overall Reaction:**



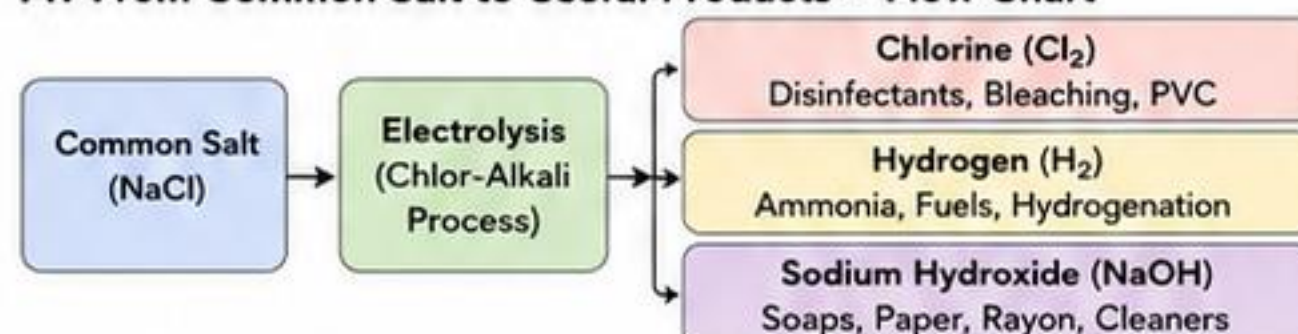
9. Electrolytic Cell for Chlor-Alkali Process (Diaphragm Cell)



10. Products of Chlor-Alkali Process and Their Uses

Product	Formula	Properties	Uses
Chlorine	Cl_2	Greenish-yellow gas, poisonous, bleaching and disinfecting property	- Disinfection of drinking water - Bleaching of paper and cloth - Manufacture of bleaching powder, PVC, HCl, etc.
Hydrogen	H_2	Colourless, odourless, burns with a pop sound	- Manufacture of ammonia (Haber's process) - Hydrogenation of oils - Fuel in rockets
Sodium Hydroxide	$NaOH$	White, hygroscopic, strongly alkaline, corrosive	- Making soaps and detergents - Paper industry - Drain cleaners - Manufacture of rayon

11. From Common Salt to Useful Products – Flow Chart



EXAM TIPS – CENTUM FOCUS

- ★ Learn definitions word by word. Write key ions: Na^+ , Cl^- , OH^- , H^+ .
- 🔑 Memorise general formulae of salt families with examples.
- ✍ Draw and label the chlor-alkali cell neatly; include anode (+) and cathode (-), diaphragm and products.
- ✅ Practice balanced equations: neutralization and chlor-alkali overall reaction.
- 💡 Answer application-based questions: sources of salt, uses of products, and advantages of electrolysis over chemical methods.

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SECTION I - COMPARISONS, COMMON MISTAKES, APPLICATIONS AND EXTRA HIGH-SCORE NOTES

1. Acid vs Base vs Salt

Feature	Acid	Base	Salt
Definition	Substances that produce H^+ ions in water.	Substances that produce OH^- ions in water.	Ionic compounds formed by reaction between an acid and a base.
Ions in Water	H^+ (Hydronium ion, H_3O^+)	OH^- (Hydroxide ion)	Depends on the salt (positive and negative ions)
Taste	Sour	Bitter	Generally salty
Feel	No feeling	Soapy / slippery	No feeling
Effect on Litmus	Blue litmus $\rightarrow$ Red	Red litmus $\rightarrow$ Blue	No change (Neutral salt)
pH Range	0 – 6	8 – 14	~ 7 (Neutral salt) or varies for other salts
Conductivity in Water	Conducts (ions present)	Conducts (ions present)	Conducts (ions present)
Formation	In water, give H^+ (or H_3O^+) ions.	In water, give OH^- ions.	By reaction between an acid and a base.
Examples	HCl, H_2SO_4 , CH_3COOH (weak acid)	NaOH, KOH, $Ca(OH)_2$, NH_4OH	NaCl, Na_2CO_3 , $CaCO_3$, NH_4Cl
Neutralisation Reaction	Acid + Base $\rightarrow$ Salt + Water (e.g., $HCl + NaOH \rightarrow NaCl + H_2O$)	—	Result of neutralisation reaction.
Effect on Metals	React with active metals to release H_2 gas.	Generally do not react with most metals.	Generally no reaction.
Examples in Daily Life	Vinegar, lemon juice, tamarind, curd	Soap, baking soda solution, toothpaste	Table salt, washing soda, baking soda

2. Strong vs Weak Acids/Bases

Feature	Strong Acids / Bases	Weak Acids / Bases
Ionisation in Water	Completely ionise to produce many ions.	Partially ionise to produce few ions.
Conductivity	High conductivity	Low conductivity
pH of Solution	Very low (acids) or very high (bases)	Moderate pH
Examples (Acids)	HCl, H_2SO_4 , HNO_3	CH_3COOH , H_2CO_3
Examples (Bases)	NaOH, KOH, $Ca(OH)_2$	NH_4OH
Reaction with Metals	React vigorously (acids) or strongly basic	React slowly (if any)
In Exams	Strong = Complete ionisation	Weak = Partial ionisation

3. Baking Soda vs Washing Soda

Feature	Baking Soda	Washing Soda
Chemical Name	Sodium hydrogen carbonate	Sodium carbonate decahydrate
Formula	$NaHCO_3$	$Na_2CO_3 \cdot 10H_2O$
Nature	Mildly basic	Strongly basic
Use	Used in baking (releases CO_2)	Used in cleaning, removing permanent hardness
Solubility in Water	Moderate	High

4. Bleaching Powder vs Washing Powder (Different!)

Feature	Bleaching Powder	Washing Powder
Chemical Name	Calcium oxychloride	Sodium carbonate (with additives)
Formula	$CaOCl_2$	$Na_2CO_3 \cdot 10H_2O$ (with STPP, etc.)
Nature	Releases Cl_2 (bleaching agent)	Cleans by removing dirt and grease
Use	Disinfects water, bleaches clothes, floors	Used for washing clothes
Note	Bleaching powder is NOT washing powder. They are different.	

5. Gypsum vs Plaster of Paris

Feature	Gypsum	Plaster of Paris
Formula	$CaSO_4 \cdot 2H_2O$	$CaSO_4 \cdot 1/2H_2O$
Name	Calcium sulphate dihydrate	Calcium sulphate hemihydrate
Nature	Hard	Powdery, sets into solid
Preparation	Found in nature	By heating gypsum at 373 K
Use	Cement, soil conditioner	Setting fractures, moulds, craftwork
Reactivity	Less reactive	More reactive

6. Hydrated Salts vs Anhydrous Salts

Feature	Hydrated Salts	Anhydrous Salts
Definition	Contain fixed number of water molecules.	Do not contain water molecules.
Formula	e.g., $CuSO_4 \cdot 5H_2O$	e.g., $CuSO_4$
Appearance	Crystals, often coloured	Usually white powder
On Heating	Lose water and become anhydrous.	No change
Example	Blue vitriol ($CuSO_4 \cdot 5H_2O$)	Anhydrous copper sulphate ($CuSO_4$)

7. Common Mistakes Students Make

- Confusing strength with concentration.
- Forgetting that water is needed for acid/base behaviour.
- Adding water to acid instead of acid to water.
- Mixing up indicator colours.
- Forgetting gas tests (H_2 , CO_2 , NH_3 , Cl_2).
- Using wrong formulas or not balancing equations.
- Not stating observation and conclusion.
- Confusing bleaching powder with washing powder.

8. How to Avoid These Mistakes in Exam

- Read the question carefully and underline key words.
- Write correct chemical formulas and balance equations.
- Remember: Acid + Water $\rightarrow$ Acid acts | Base + Water $\rightarrow$ Base acts
- Always add acid to water, never water to acid.
- Memorise indicator colours using tables.
- Learn gas tests with observations.
- Write steps neatly and give clear conclusion.
- Revise common pairs: baking soda vs washing soda, gypsum vs POP, etc.

9. Real-Life Applications of Acids, Bases and Salts

Field	Applications
Food	Acetic acid (vinegar) – preservative; Citric acid – flavouring; Baking soda – baking.
Medicine	Antacids (neutralise stomach acid); Aspirin contains carboxylic acid.
Agriculture	Calcium carbonate – neutralises acidic soil; Ammonia – fertiliser.
Household Cleaning	Bases in soaps remove grease; Baking soda cleans and deodorises.
Industries	H_2SO_4 in battery, fertiliser and dye industries; NaOH in paper and soap making.
Water Treatment	Alum ($KAl(SO_4)_2 \cdot 12H_2O$) used to purify drinking water.
Fire Safety	Sodium bicarbonate used in fire extinguishers (releases CO_2).
Construction	Cement contains calcium compounds; Plaster of Paris used for finishing.

10. Short Assertion-Reason Style Facts (True/False Type)

1. Assertion: All bases are slippery. Reason: Bases feel soapy.
2. Assertion: HCl is a strong acid. Reason: It ionises completely in water.
3. Assertion: Na_2CO_3 is a salt. Reason: It is formed by acid-base reaction.
4. Assertion: Baking soda is more basic than washing soda. Reason: Washing soda is strongly basic.
5. Assertion: Gypsum loses water to form Plaster of Paris. Reason: Heating gypsum removes water.

11. One-Mark Fact Bank (Quick Recall)

- Litmus turns red in acid and blue in base.
- Neutral salts do not change the colour of litmus.
- $HCl + NaOH \rightarrow NaCl + H_2O$ is a neutralisation reaction.
- The pH of acid is less than 7 and base is more than 7.
- Plaster of Paris sets into hard solid when water is added.
- Bleaching powder releases chlorine gas in moist air.

12. Daily-Life Examples (At a Glance)

Lemon Juice (citric acid)  Sour, acidic.	Curd  Contains lactic acid.	Vinegar (acetic acid)  Acidic, used in pickles.	Baking Soda ($NaHCO_3$)  Used in baking, relieves acidity.	Soap (basic)  Made from NaOH/KOH.	Washing Soda ($Na_2CO_3 \cdot 10H_2O$)  Removes permanent hardness.	Table Salt (NaCl)  Used in food, preservative.	Bleaching Powder ($CaOCl_2$)  Disinfects water, bleaches.	Gypsum ($CaSO_4 \cdot 2H_2O$)  Used in cement and soil.
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EXAM TIPS – CENTUM FOCUS

- ★ Make comparison tables in your notebook – easy to score in long answers.
- ✏ Learn difference pairs (Baking soda vs Washing soda, Gypsum vs POP, etc.) – common in exams.
- ✍ Write balanced equations and mention physical changes (colour change, gas evolution, precipitate).
- ✓ Link chapter with daily life examples – helps in both short and long answers.
- 📌 Practice NCERT back exercises, diagrams (indicator changes, gas tests) and PYQs.

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SECTION F - WASHING SODA, PLASTER OF PARIS AND WATER OF CRYSTALLISATION

1. Washing Soda

- Chemical name: Sodium carbonate decahydrate ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$).
- What it is: A white crystalline solid.
- It is also called soda crystals.
- It is more soluble in water than baking soda.

Properties

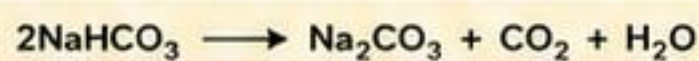
- White, transparent crystals.
- Soluble in water.
- Alkaline in nature (basic).
- Loses water of crystallisation on heating.

Uses

- Used in glass, soap and paper industries.
- Used in cleaning agents and water softening.
- Used in removal of permanent hardness of water.

2. Preparation of Washing Soda From Baking Soda

- When baking soda (sodium hydrogen carbonate) is heated, it gives sodium carbonate.



- On dissolving the sodium carbonate in water and cooling, washing soda crystallises out.



- The crystals are separated by filtration and dried. These are washing soda crystals.

3. Washing Soda Removes Permanent Hardness of Water

Hard water (contains Ca^{2+} , Mg^{2+})

After adding washing soda

CaCO_3
 MgCO_3 (ppt.)

Washing soda precipitates Ca^{2+} and Mg^{2+} ions as insoluble carbonates, thus removing permanent hardness.

4. Water of Crystallisation

- Some salts contain a fixed number of water molecules in their crystals. This water is called **water of crystallisation**.

Significance

- It affects the mass, colour, physical properties and uses of salts.
- It is lost on heating and regained on adding water.

Hydrated Salts (Contain water of crystallisation)	Anhydrous Salts (No water of crystallisation)
- Have water molecules in their crystal structure. - Written with water molecules in formula, e.g., $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$.	- Do not contain water in their crystal structure. - Written without water molecules, e.g., CuSO_4.

5. Examples of Hydrated Salts

Salt	Formula	Number of Water Molecules	Colour of Hydrated Salt
Copper sulphate	$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$	5	Blue
Calcium sulphate	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$	2	White
Ferrous sulphate	$\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$	7	Pale green
Washing soda	$\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$	10	White

6. Copper Sulphate Crystals – Colour Change

On Heating (Loses Water of Crystallisation)

Blue crystals ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$) $\xrightarrow{\text{Heat}}$ White powder (CuSO_4) + $5\text{H}_2\text{O}$ (Water vapour)

$$\text{CuSO}_4 \cdot 5\text{H}_2\text{O} \longrightarrow \text{CuSO}_4 + 5\text{H}_2\text{O}$$

On Adding Water (Regains Water of Crystallisation)

White powder (CuSO_4) $\xrightarrow{\text{Add water}}$ Blue crystals ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$)

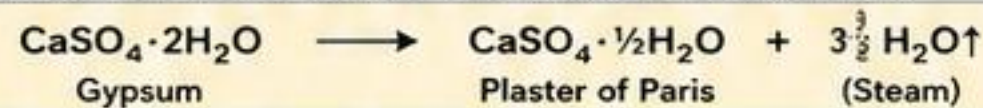
$$\text{CuSO}_4 + 5\text{H}_2\text{O} \longrightarrow \text{CuSO}_4 \cdot 5\text{H}_2\text{O}$$

7. Gypsum and Plaster of Paris

- Gypsum: Calcium sulphate dihydrate ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$).
- Plaster of Paris (POP): Calcium sulphate hemihydrate ($\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$).
- Both are white solids.

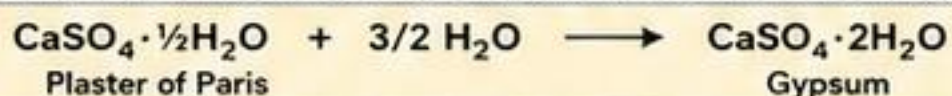
8. Preparation of Plaster of Paris (POP)

- Gypsum is heated strongly to 373 K (100 °C) to form POP.



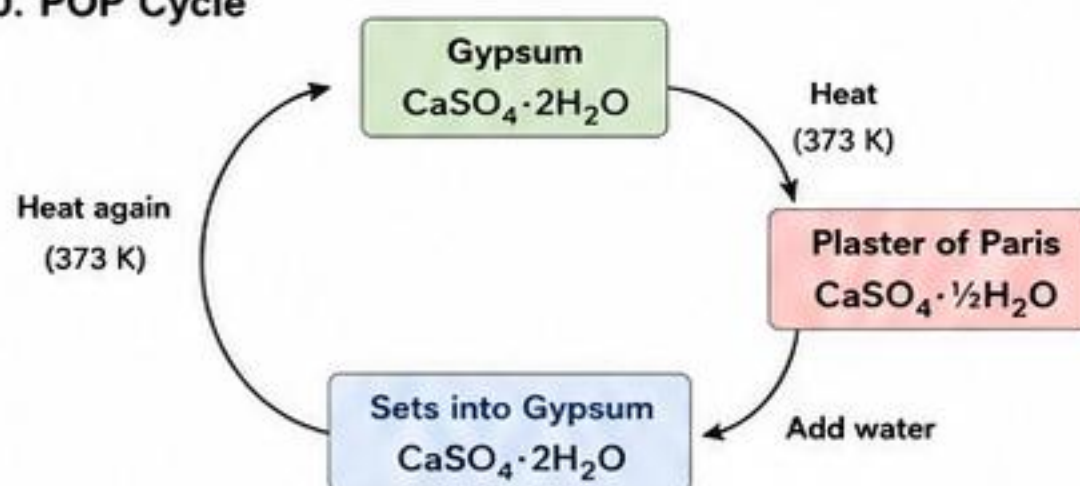
9. Setting of Plaster of Paris

- When POP is mixed with water, it sets into a hard mass of gypsum.



- This is an exothermic reaction (gives out heat).

10. POP Cycle



11. Comparison – Baking Soda vs Washing Soda

Feature	Baking Soda	Washing Soda
Chemical Name	Sodium hydrogen carbonate	Sodium carbonate decahydrate
Formula	NaHCO_3	$\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$
Nature	Mildly basic	Strongly basic
Solubility in Water	Less soluble	More soluble
Uses	Baking, medicine, antacid	Cleaning, industries, water softening

12. Comparison – Gypsum vs Plaster of Paris (POP)

Feature	Gypsum	Plaster of Paris (POP)
Formula	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$	$\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$
Water of Crystallisation	2 water molecules	$\frac{1}{2}$ water molecule
Preparation	Naturally occurring	By heating gypsum
When mixed with water	No change	Sets into hard mass (gypsum)
Uses	Cement, fertilizers	Moulds, casts, sculptures, bandages

EXAM TIPS – CENTUM FOCUS

- ★ List uses of washing soda and POP.
- ✏ Learn equations for preparation and setting of POP.
- ✓ Remember water of crystallisation and examples with colours.
- 🎯 Practice diagram-based questions (hardness removal, POP cycle).

- 📌 Write balanced equations correctly.
- 💧 Note the change in colour of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ crystals.
- 📖 Compare properties and uses for 2-mark answers.

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SECTION G - COMPLETE EQUATION BANK WITH OBSERVATIONS, COLOUR CHANGES AND USES

S. No.	Balanced Equation	Type/Concept	Observation	Colour/Inference	Use/Key Point
1.	$\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$	Neutralisation (Strong acid + Strong base)	Heat evolved.	No colour change.	Antacid action; neutralisation.
2.	$\text{H}_2\text{SO}_4 + 2\text{KOH} \rightarrow \text{K}_2\text{SO}_4 + 2\text{H}_2\text{O}$	Neutralisation	Heat evolved.	No colour change.	Used in fertilizers and industrial processes.
3.	$\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\uparrow$	Metal + Acid (displacement)	Effervescence; gas burns with 'pop' sound.	Colourless gas (H_2). Zn dissolves.	Preparation of H_2 gas; reactivity of metals.
4.	$\text{Fe} + 2\text{HCl} \rightarrow \text{FeCl}_2 + \text{H}_2\uparrow$	Metal + Acid (displacement)	Effervescence; gas burns with 'pop' sound.	Solution turns pale green.	Preparation of ferrous salts; reactivity.
5.	$\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2\uparrow$	Acid + Carbonate	Brisk effervescence.	CO_2 is colourless.	CO_2 used in soda-acid fire extinguishers.
6.	$\text{NaHCO}_3 + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2\uparrow$	Acid + Bicarbonate	Effervescence.	CO_2 is colourless.	Baking soda neutralises acids; antacid.
7.	$\text{CuO} + 2\text{HCl} \rightarrow \text{CuCl}_2 + \text{H}_2\text{O}$	Acid + Basic Oxide	Black solid dissolves.	Solution turns green.	Removal of basic oxide layers; lab preparation.
8.	$\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3\downarrow + \text{H}_2\text{O}$	Base + Acidic Oxide (Precipitation)	White precipitate forms.	White CaCO_3 ppt.	Limewater test for CO_2 .
9.	$2\text{NaOH} + \text{CO}_2 \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$	Base + Acidic Oxide	No visible change (initially).	No colour change.	Absorption of CO_2 in alkaline solutions.
10.	$2\text{NaCl} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{Cl}_2\uparrow + \text{H}_2\uparrow$	Electrolysis of Brine	Bubbles of Cl_2 and H_2 at electrodes.	Cl_2 is greenish-yellow gas; H_2 colourless.	Industrial production of NaOH , Cl_2 and H_2 .
11.	$\text{Ca(OH)}_2 + \text{Cl}_2 \rightarrow \text{CaOCl}_2 + \text{H}_2\text{O}$	Bleaching Powder Formation	Chlorine absorbed; slight rise in temp.	Product is white.	Bleaching and disinfecting agent.
12.	$2\text{NaHCO}_3 \rightarrow \text{Na}_2\text{CO}_3 + \text{CO}_2\uparrow + \text{H}_2\text{O}$	Thermal Decomposition (Bicarbonate)	Effervescence; gas evolved.	CO_2 is colourless.	Baking, laboratory preparations.
13.	$\text{Na}_2\text{CO}_3 + 10\text{H}_2\text{O} \rightarrow \text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$	Crystallisation (Decahydrate)	White crystals form on cooling.	Sodium carbonate decahydrate.	Washing soda manufacture.
14.	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O} \rightarrow \text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O} + \frac{3}{2}\text{H}_2\text{O}$	Dehydration (Gypsum to POP)	Water driven off on heating.	White powder obtained.	Plaster of Paris (POP) preparation.
15.	$\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O} + \frac{3}{2}\text{H}_2\text{O} \rightarrow \text{CaSO}_4 \cdot 2\text{H}_2\text{O}$	Hydration (POP to Gypsum)	POP sets hard in water.	Hard mass forms.	Castings, moulds, bandages.
16.	$\text{CuSO}_4 \cdot 5\text{H}_2\text{O} \rightarrow \text{CuSO}_4 + 5\text{H}_2\text{O}$	Dehydration of Hydrated Salt	Blue crystals lose water on heating.	Blue $\rightarrow$ White.	Water of crystallisation removal.
17.	$\text{CuSO}_4 + 5\text{H}_2\text{O} \rightarrow \text{CuSO}_4 \cdot 5\text{H}_2\text{O}$	Hydration of Anhydrous Salt	Water absorbed; crystals form.	White $\rightarrow$ Blue crystals.	Preparation of hydrated copper(II) sulfate.
18.	$\text{NH}_3 + \text{HCl} \rightarrow \text{NH}_4\text{Cl}$	Acid + Base (Gas Neutralisation)	White fumes/form white solid.	White ammonium chloride.	Laboratory preparation of NH_4Cl .

A. COMMON INDICATOR COLOUR CHANGES

Indicator	In Acidic Solution	In Neutral Solution	In Basic Solution
Litmus	Red	Purple	Blue
Phenolphthalein	Colourless	Colourless	Pink
Methyl orange	Red	Orange	Yellow
Turmeric (natural)	Yellow	Yellow	Reddish brown
China rose (natural)	Pink	Pink	Green

B. MUST MEMORISE FORMULAS (FROM CHAPTER 2)

- Water $\rightarrow \text{H}_2\text{O}$
- Sodium hydroxide $\rightarrow \text{NaOH}$
- Hydrochloric acid $\rightarrow \text{HCl}$
- Sulphuric acid $\rightarrow \text{H}_2\text{SO}_4$
- Sodium chloride $\rightarrow \text{NaCl}$
- Carbon dioxide $\rightarrow \text{CO}_2$
- Calcium carbonate $\rightarrow \text{CaCO}_3$
- Sodium carbonate $\rightarrow \text{Na}_2\text{CO}_3$
- Sodium hydrogen carbonate $\rightarrow \text{NaHCO}_3$
- Calcium hydroxide $\rightarrow \text{Ca(OH)}_2$
- Plaster of Paris $\rightarrow \text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$
- Gypsum $\rightarrow \text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
- Copper(II) sulfate (anhydrous) $\rightarrow \text{CuSO}_4$
- Copper(II) sulfate pentahydrate $\rightarrow \text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
- Bleaching powder $\rightarrow \text{CaOCl}_2$

C. COMMON OBSERVATION PHRASES (FOR EXAM)

- Brisk effervescence with evolution of a colourless gas.
- The gas turns limewater milky.
- The blue litmus turns red.
- The red litmus remains red.
- Heat is evolved.
- The solution turns from blue to green.
- A white precipitate is formed.
- The solid dissolves to form a colourless / coloured solution.
- Greenish-yellow gas is evolved.
- The blue crystals turn white on heating.
- White solid absorbs water and turns hard.

D. WHICH SALT/CHEMICAL IS USED FOR WHAT?

Salt/Chemical	Use
Sodium carbonate (Na_2CO_3)	Washing soda; glass, soap and detergent manufacture.
Sodium hydrogen carbonate (NaHCO_3)	Antacid; baking soda; in fire extinguishers.
Calcium carbonate (CaCO_3)	Making cement, glass; antacid.
Bleaching powder (CaOCl_2)	Bleaching cotton and wood pulp; disinfectant.
Plaster of Paris ($\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$)	Moulds, casts, false ceilings, bandages.
Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$)	Cement manufacture; soil conditioner.
Copper(II) sulfate ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$)	Fungicide (Bordeaux mixture); lab reagent.

EXAM TIPS – CENTUM FOCUS

- ★ Learn all equations with states (s), (l), (g), (aq) and $\uparrow$ for gas, $\downarrow$ for precipitate.
- 🔗 Link observation with inference (e.g., limewater turns milky $\rightarrow \text{CO}_2$).
- ✍ Write balanced equations neatly – full marks depend on correctness.
- 🧠 Revise indicator colours and uses – frequently asked in theory + case-based questions.

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SECTION H - IMPORTANT DIAGRAMS, LAB TESTS AND VISUAL RECAP

1. Acid in Water and Base in Water – Ion Formation

HCl in Water (Acid)

$$\text{HCl} + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{Cl}^-$$

HCl (Acid) + H₂O (Water) → H₃O⁺ (Hydronium ion) + Cl⁻ (Chloride ion)

NaOH in Water (Base)

$$\text{NaOH} \rightarrow \text{Na}^+ + \text{OH}^-$$

NaOH (Base) → Na⁺ (Sodium ion) + OH⁻ (Hydroxide ion)

2. Zinc + Dilute HCl – Hydrogen Gas and Pop Test

Zn granules + Dilute HCl → H₂ gas

'Pop' sound

Burning splint

- $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2 \uparrow$
- Hydrogen burns with a 'pop' sound.

3. Acid + Carbonate → CO₂ Passed Through Limewater

Dilute HCl + Na₂CO₃ (Sodium carbonate) → CO₂ gas

Limewater Ca(OH)₂

Clear → Milky (CaCO₃)

- $\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2 \uparrow$
- CO₂ turns limewater milky due to CaCO₃.

4. Chlor-Alkali Process – Electrolytic Cell

DC Power Supply

Anode (Graphite): $\text{Cl}^- \rightarrow \text{Cl}_2 + \text{e}^-$ (at anode)

Cathode (Iron): $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$ (at cathode)

Brine (NaCl solution)

Na⁺, OH⁻ in solution

- Products:** Cl₂ (at anode), H₂ (at cathode), NaOH (in solution)
- Overall:** $2\text{NaCl} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2 + \text{Cl}_2$

5. Preparation of Bleaching Powder

Cl₂ (Chlorine gas) + Dry slaked lime [Ca(OH)₂] (Powder) → Bleaching powder (Calcium oxychloride) CaOCl₂

$$\text{Ca(OH)}_2 + \text{Cl}_2 \rightarrow \text{CaOCl}_2 + \text{H}_2\text{O}$$

Uses: Disinfecting drinking water, bleaching cotton/linen, as an oxidising agent.

6. Cake / Bread Rising – Baking Powder and CO₂

Baking powder (NaHCO₃) → Heat / Moisture → CO₂ gas (Bubbles trapped in dough/batter) → Cake rises and becomes soft & spongy.

$$2\text{NaHCO}_3 \xrightarrow{\text{Heat}} \text{Na}_2\text{CO}_3 + \text{CO}_2 \uparrow + \text{H}_2\text{O}$$

7. Fire Extinguisher (Soda Acid Type)

Dilute H₂SO₄ (Acid) + NaHCO₃ (Baking soda) solution → CO₂ gas released

- On opening, acid + NaHCO₃ → CO₂ gas.
- CO₂ smothers fire by cutting off supply of air.

8. Washing Soda – Removes Permanent Hardness

Hard water (Ca²⁺ / Mg²⁺ ions present) + Add Washing soda (Na₂CO₃) → Soft water (Hardness removed)

Reactions:

$$\text{CaSO}_4 + \text{Na}_2\text{CO}_3 \rightarrow \text{CaCO}_3 \downarrow + \text{Na}_2\text{SO}_4$$

$$\text{MgSO}_4 + \text{Na}_2\text{CO}_3 \rightarrow \text{MgCO}_3 \downarrow + \text{Na}_2\text{SO}_4$$

(Insoluble carbonates removed by filtration)

9. Copper Sulphate – Hydration / Dehydration

Blue crystals CuSO₄ · 5H₂O (Hydrated) → Heat → White powder CuSO₄ (Anhydrous)

White powder CuSO₄ (Anhydrous) → Add water → Blue again

- Heating removes water → white anhydrous CuSO₄.
- Adding water brings back blue colour.

10. Gypsum → POP → Gypsum Cycle

Gypsum CaSO₄ · 2H₂O → Heat (373 K) → Plaster of Paris (POP) CaSO₄ · ½H₂O → Add water → Gypsum CaSO₄ · 2H₂O

- POP + water → sets hard to gypsum.
- Used in casts, moulds, false ceilings, toys, etc.

11. pH Scale (Universal Indicator)

Acidic (0-6) | Neutral (7) | Basic (8-14)

pH < 7 Acidic | pH = 7 Neutral | pH > 7 Basic

- Lower pH → stronger acid.
- Higher pH → stronger base.

12. Indicator Colour Changes (Mini Table)

Indicator	Acidic Solution	Basic Solution	Neutral Solution
Litmus	Red	Blue	Purple
Phenolphthalein	Colourless	Pink	Colourless
Methyl orange	Red	Yellow	Orange
Turmeric (natural)	Yellow	Reddish brown	Yellow
Universal indicator	Red-Orange (0-6)	Green (7)	Blue-Violet (8-14)

HOW TO DRAW DIAGRAMS NEATLY IN EXAM

- Use clear, single lines and proper labels.
 - Draw diagrams with pencil first, then darken.
 - Show direction with arrows.
 - Keep diagrams proportional and not too big.
 - Label all parts and write underlined title.
- ★ Tip: Neat diagrams fetch full marks!



DIAGRAM-BASED QUESTIONS (PRACTICE)

- What gas is collected in test tube in diagram 2? How do you test it?
- What change is seen in limewater in diagram 3 and why?
- What products are obtained in the chlor-alkali process?
- Why does the cake rise in diagram 6?
- Why does washing soda remove hardness?
- What happens when blue CuSO₄ · 5H₂O is heated strongly?



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SECTION E – BLEACHING POWDER AND BAKING SODA

1. Bleaching Powder (Calcium Oxychloride)

- Bleaching powder is calcium oxychloride.
- Formula:** CaOCl_2
- It is a white or pale yellow powder with a strong chlorine smell.

2. Properties of Bleaching Powder

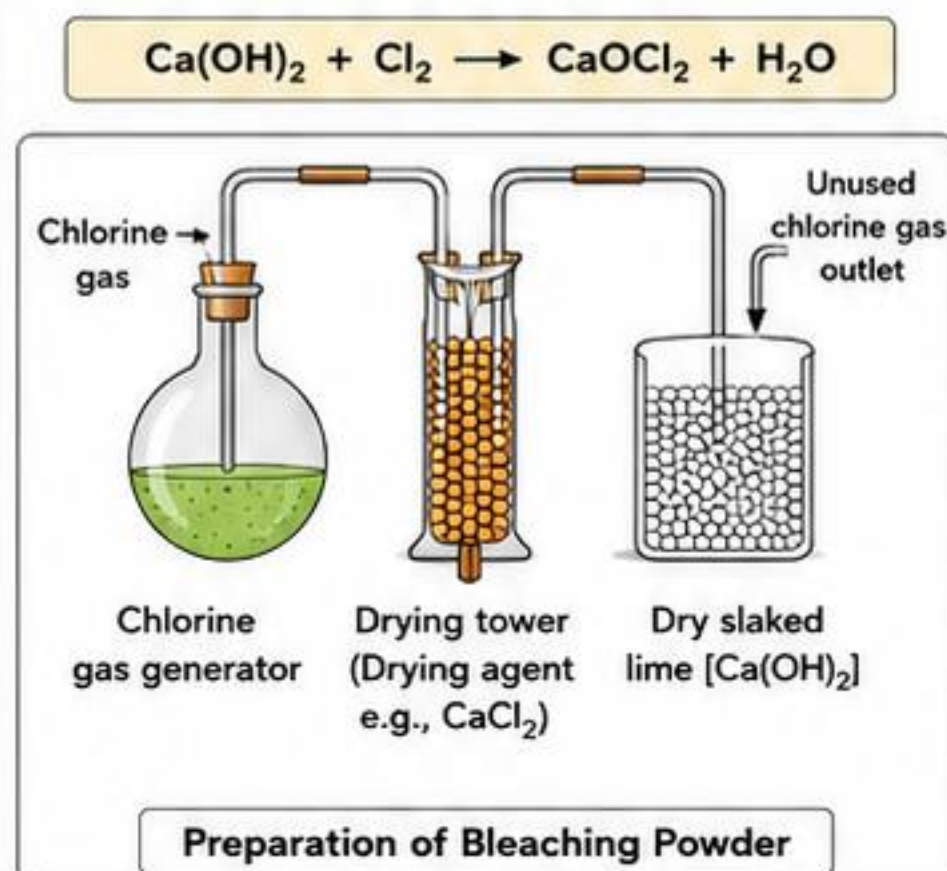
- White or pale yellow powder.
- Strong chlorine odour.
- Slightly soluble in water.
- Reacts with acids to liberate chlorine gas.
- Slowly decomposes on exposure to air and moisture.

3. Uses of Bleaching Powder

- Used as a bleaching agent for cotton, linen and wood pulp.
- Used to disinfect drinking water and wells.
- Used as an oxidising agent in many chemical industries.
- Used in the purification of sugar.

4. Preparation of Bleaching Powder

Bleaching powder is prepared by passing dry chlorine gas over dry slaked lime.



5. Why Bleaching Powder is Stored Airtight?

- It decomposes on exposure to air and moisture.
- Chlorine gas is released and its bleaching property decreases.
- Therefore, it is stored in airtight containers.

6. Observations and Inferences (Bleaching Powder)

Observation	Inference
Turns moist blue litmus red.	It has basic nature.
On adding dilute acid, a gas with pungent smell is evolved.	Chlorine gas is liberated.
With KI–starch paper, blue colour appears.	Chlorine gas is confirmed.

7. Baking Soda (Sodium Hydrogencarbonate)

- Baking soda is sodium hydrogencarbonate.
- Formula:** NaHCO_3
- It is a white crystalline powder.

8. Properties of Baking Soda

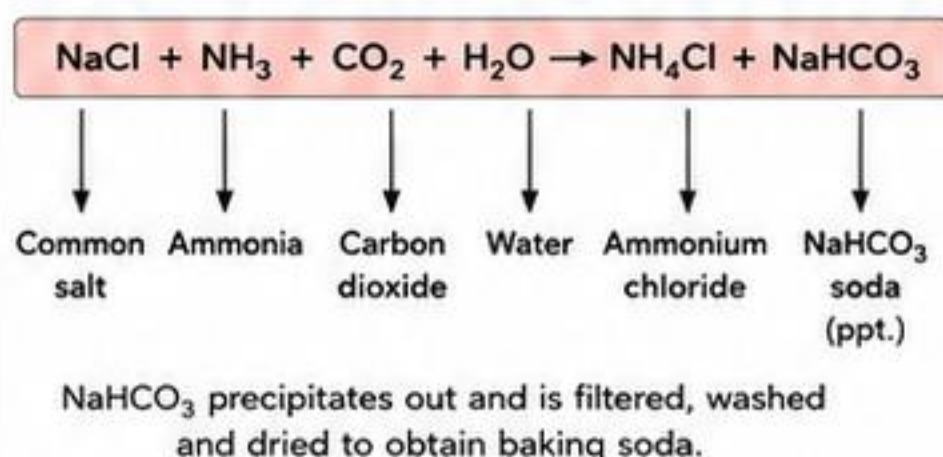
- White crystalline solid.
- Slightly soluble in water.
- Mildly basic in nature.
- On heating, it decomposes to give sodium carbonate, carbon dioxide and water.

9. Uses of Baking Soda

- Used in making baking powder.
- Used as an antacid to relieve acidity.
- Used in soda–acid fire extinguishers.
- Used in the manufacture of glass, paper and soap.

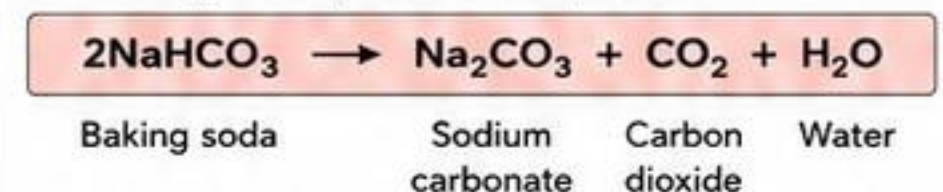
10. Preparation of Baking Soda (Solvay Route – Simplified)

Baking soda is prepared by the Solvay process using common salt.



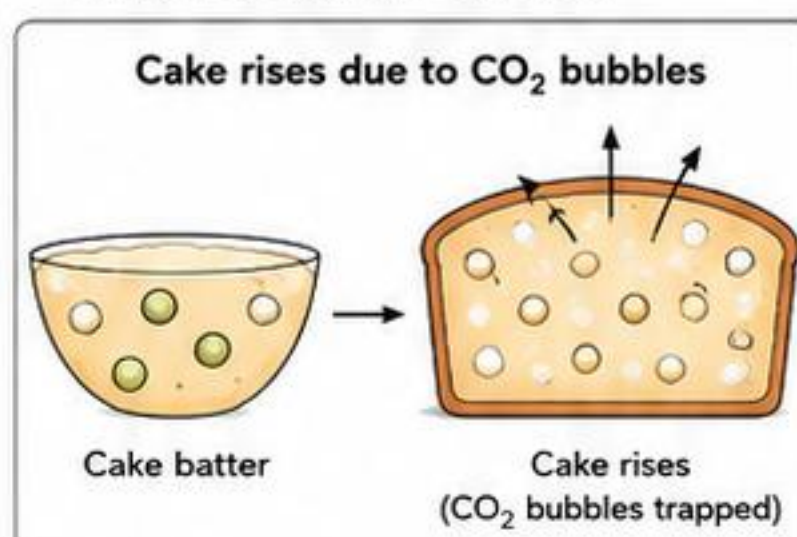
11. Heating of Baking Soda

On heating, baking soda decomposes.



12. Baking Powder

- Baking powder is a mixture of baking soda and a mild edible acid (e.g., tartaric acid).
- When mixed with water in the batter, acid reacts with baking soda to release CO_2 gas.
- CO_2 gets trapped in the batter, making it soft and spongy. The cake rises.



13. Baking Soda in Fire Extinguishers

- In soda–acid type fire extinguishers, baking soda (NaHCO_3) is used.
- When acid (H_2SO_4) is added, CO_2 gas is released which puts out the fire by cutting off the air supply.



Soda–Acid Fire Extinguisher



14. Bleaching Powder vs Baking Soda

Feature	Bleaching Powder (CaOCl_2)	Baking Soda (NaHCO_3)
Chemical Name	Calcium oxychloride	Sodium hydrogencarbonate
Formula	CaOCl_2	NaHCO_3
Nature	Basic	Mildly basic
Appearance	White / pale yellow powder	White crystalline solid
Solubility	Slightly soluble in water	Slightly soluble in water
Odour	Strong chlorine smell	No smell
Preparation	Dry Cl_2 passed over dry slaked lime	Solvay process using common salt
Main Use	Bleaching, disinfecting, purifying water	Baking, antacid, fire extinguishers
Special Property	Liberates Cl_2 with acids	Releases CO_2 on heating or with acids

EXAM TIPS – CENTUM FOCUS

- ★ Learn names and formulae by heart.
- 🔑 Draw preparation diagrams neatly and label all parts.
- ✍ Write balanced chemical equations with state symbols where needed.
- ✅ Compare and contrast in tables for better marks.
- 🔍 Practice short answers, equations and labelled diagrams.

MUST MEMORIZE

- ✓ Bleaching powder – CaOCl_2 – prepared by passing Cl_2 over dry slaked lime.
- ✓ Bleaching powder reacts with acids to liberate Cl_2 .
- ✓ Baking soda – NaHCO_3 – used in baking, antacids and fire extinguishers.
- ✓ Heating NaHCO_3 – $2\text{NaHCO}_3 \rightarrow \text{Na}_2\text{CO}_3 + \text{CO}_2 + \text{H}_2\text{O}$
- ✓ Soda–acid fire extinguisher – $2\text{NaHCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + 2\text{CO}_2 + 2\text{H}_2\text{O}$

Ch 2: Acids, Bases and Salts

Complete Notes | Deep Definitions | Diagrams | Tables | Exam Tips | Centum Focus

SECTION J - HIGH-SCORE REVISION SHEET

1. MUST-KNOW DEFINITIONS (WITH EXAMPLE)

Term	Definition	One Example
Acid	A substance that turns blue litmus red. In water, it produces H^+ (H_3O^+) ions.	Hydrochloric acid (HCl)
Base	A substance that turns red litmus blue. In water, it produces OH^- ions.	Sodium hydroxide (NaOH)
Salt	A substance formed when an acid reacts with a base (neutralization).	Sodium chloride (NaCl)
Indicator	A substance that shows different colours in acidic and basic solutions.	Phenolphthalein
pH	A number that shows how acidic or basic a solution is (0–14 scale).	Lemon juice pH $\approx$ 2
Neutralization	Reaction between an acid and a base to form salt and water.	$HCl + NaOH \rightarrow NaCl + H_2O$
Strong Acid	An acid that ionizes completely in water.	HCl, H_2SO_4
Weak Acid	An acid that ionizes partially in water.	CH_3COOH (Acetic acid)
Strong Base	A base that dissociates completely in water.	NaOH, KOH
Weak Base	A base that dissociates partially in water.	NH_4OH (Ammonia)
Water of Crystallisation	Fixed number of water molecules present in a salt crystal.	$CuSO_4 \cdot 5H_2O$ (Blue vitriol)

2. COMMON FACTS & QUICK RULES

- Acids taste sour. Bases taste bitter and feel soapy.
- All acids release H^+ ions in water.
- All bases release OH^- ions in water.
- Neutral solution has pH = 7.
- pH < 7 $\rightarrow$ Acidic; pH > 7 $\rightarrow$ Basic; pH = 7 $\rightarrow$ Neutral.
- Dilution of acid/base: add acid/base to water, not water to acid/base.
- Stronger acid = more H^+ ions; Stronger base = more OH^- ions.
- Reaction of acid + metal $\rightarrow$ salt + hydrogen gas.
- Reaction of acid + carbonate/bicarbonate $\rightarrow$ salt + CO_2 + H_2O .
- Indicators: turn red in acid, blue in base (litmus).
- Phenolphthalein: colourless (acid), pink (base).
- Neutralization always produces salt and water.
- Water of crystallisation is lost on heating.

3. COMMON EXAM MISTAKES & HOW TO AVOID

Common Mistake	How to Avoid
Writing H_2O instead of H_3O^+ for acid in water.	Always write H_3O^+ in aqueous acid.
Forgetting to balance equations.	Balance atoms on both sides.
Confusing strong and weak acids/bases.	Memorize examples and practice.
Wrong indicator colours.	Learn table: litmus & phenolphthalein.
Ignoring states (aq), (s), (g).	Write physical states in equations.
Forgetting CO_2 in carbonate reactions.	Acid + carbonate $\rightarrow CO_2$ + H_2O + salt.
Not using correct formula of salts.	Learn formulas: NaOH, Na_2CO_3 , $CaSO_4$, etc.
Mixing up baking soda and washing soda.	Baking soda = $NaHCO_3$; Washing soda = $Na_2CO_3 \cdot 10H_2O$.

4. COMPACT COMPARISON TABLES

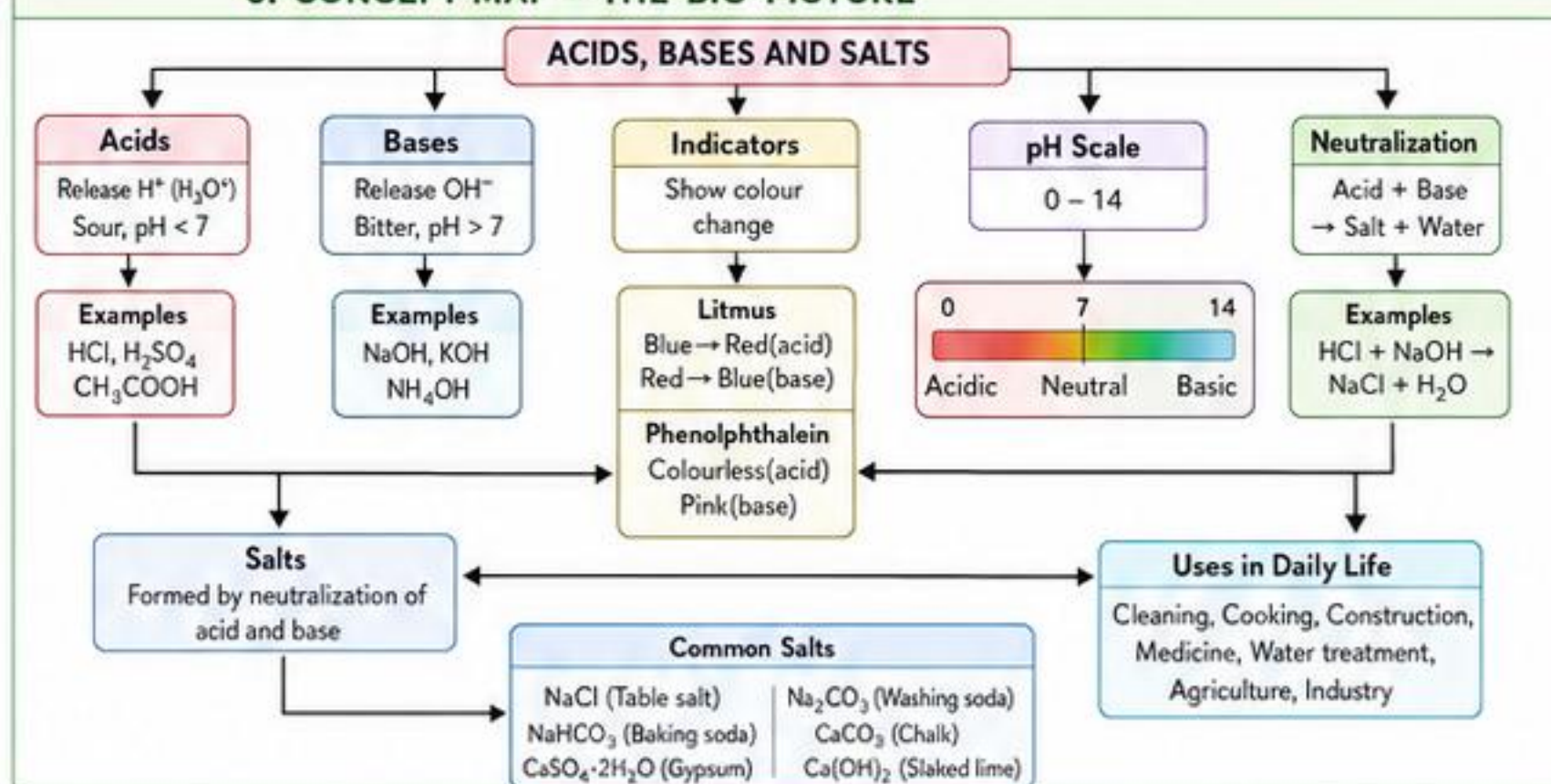
Acid vs Base			Strong vs Weak		
Feature	Acid	Base	Feature	Strong	Weak
Taste	Sour	Bitter	Ionization	Complete	Partial
Litmus	Blue $\rightarrow$ Red	Red $\rightarrow$ Blue	H^+ / OH^- ions	More	Less
Ions in water	H^+ / H_3O^+	OH^-	Conductivity	High	Low
pH	< 7	> 7	pH (Same conc.)	Lower (acid)	Higher
Nature	Corrosive	Caustic	Examples (Acid)	HCl, H_2SO_4	CH_3COOH
Example	HCl	NaOH	Examples (Base)	NaOH, KOH	NH_4OH

Baking Soda vs Washing Soda			Gypsum vs POP		
Feature	Baking Soda	Washing Soda	Feature	Gypsum	POP
Formula	$NaHCO_3$	$Na_2CO_3 \cdot 10H_2O$	Formula	$CaSO_4 \cdot 2H_2O$	$CaSO_4 \cdot \frac{1}{2}H_2O$
Name	Sodium hydrogencarbonate	Sodium carbonate decahydrate	Name	Calcium sulphate dihydrate	Plaster of Paris
Nature	Slightly basic	Strongly basic	Nature	Hard	Powder
Solubility	Moderate	High	On heating	Loses $1\frac{1}{2} H_2O$	Sets to hard mass with water
On heating	Decomposes	Loses water	Use	Cement, plaster, agriculture	Cast, moulds, dental, toys
Use	Antacid, baking	Cleaning, softening hard water			

5. IMPORTANT USES – ONE LINERS

- Bleaching Powder ($CaOCl_2$):** Used for bleaching cotton/linen and for disinfecting drinking water.
- Baking Soda ($NaHCO_3$):** Used as an antacid, in baking powder and in soda-acid fire extinguishers.
- Washing Soda ($Na_2CO_3 \cdot 10H_2O$):** Used for cleaning, removing permanent hardness of water, in glass and soap making.
- POP ($CaSO_4 \cdot \frac{1}{2}H_2O$):** Used for making casts, moulds, toys, and false ceilings.
- Sodium Hydroxide (NaOH):** Used in soap making, paper industry, cleaning drains.
- Chlorine (Cl_2):** Used as a disinfectant in water treatment, and in the manufacture of bleaching powder and PVC.
- Hydrogen (H_2):** Used in hydrogenation of oils, as a fuel and in the manufacture of ammonia.

6. CONCEPT MAP – THE BIG PICTURE



7. LAST-MINUTE CHECKLIST

- Can I define acid, base, salt, indicator, pH and neutralization?
- Do I know litmus and phenolphthalein colours?
- Can I write and balance equations for acid-metal, acid-carbonate and neutralization?
- Do I remember formulas of common salts?
- Do I know the difference between strong/weak acids and bases?
- Have I memorized uses of important substances?
- Have I practiced at least 5 numerical/long-answer questions?
- Am I clear about diagrams and observations?
- Have I revised all tables and rules once?

8. MINI-SUMMARY – AT A GLANCE

Litmus Test	Phenolphthalein Test	Acid + Metal	Acid + Carbonate	Neutralization
Acid: Blue $\rightarrow$ Red Base: Red $\rightarrow$ Blue	Acid: Colourless Base: Pink	$Zn + 2HCl \rightarrow ZnCl_2 + H_2 \uparrow$ Hydrogen gas (Effervescence)	$CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 \uparrow + H_2O$ CO_2 gas (Effervescence)	$HCl + NaOH \rightarrow NaCl + H_2O$

9. CENTUM TIPS – SCORE 100%

- Concept First**
Understand concepts clearly. Don't just mug examples.
- Practice Writing**
Write definitions, equations and uses regularly. Presentation matters.
- Use Tables & Charts**
Reproduce tables and flowcharts in answers for extra marks.
- Keywords Matter**
Underline key terms and use scientific language accurately.
- Revision is Power**
Revise smartly, solve past papers and manage time in the exam.

REMEMBER EVERY DAY!

- ★ Revise a little every day – big result on exam day!
- ★ Write neat, labelled diagrams.
- ★ Read questions carefully.
- ★ Manage time – attempt all questions.
- ★ Stay calm, confident and positive!

Ch 2: Acids, Bases and Salts

Complete Notes | Deep Definitions | Diagrams | Tables | Exam Tips | Centum Focus

SECTION K - DETAILED LAB TESTS, OBSERVATIONS AND IDENTIFICATION OF UNKNOWN SOLUTIONS

In the laboratory, we can identify whether a given solution is acidic, basic or neutral using indicators and simple chemical tests. We can also identify salts by their characteristic reactions.

K1. USING INDICATORS

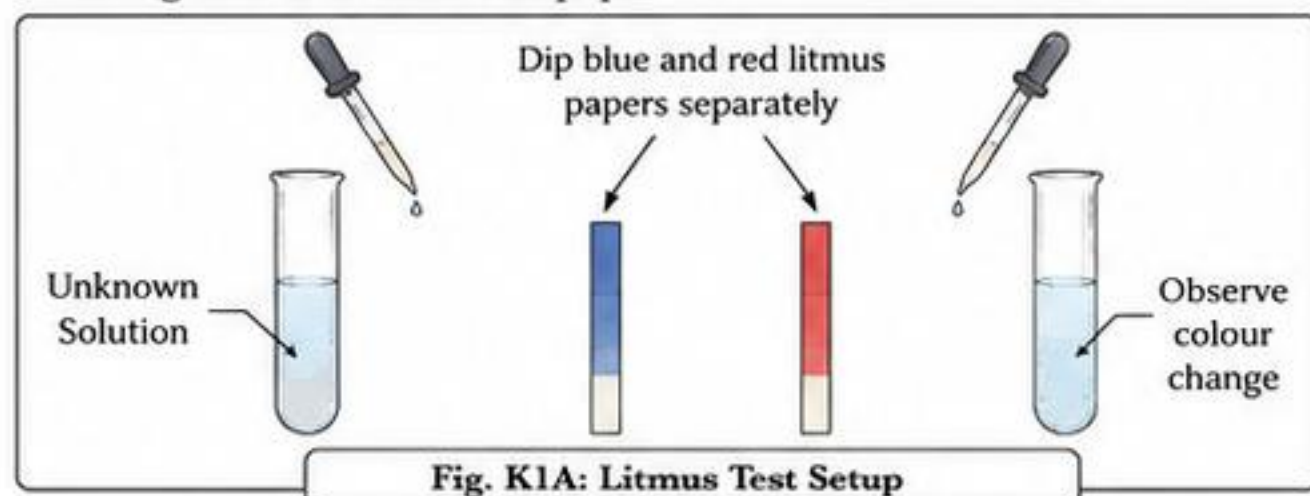
Indicators are substances that change colour in acidic or basic solutions.

➤ Common indicators used in the laboratory:

Indicator	Acidic Solution (pH < 7)	Neutral Solution (pH = 7)	Basic Solution (pH > 7)	pH Range/ Transition Range
Litmus	Blue → Red	No change (Purple)	Red → Blue	—
Phenolphthalein (Colourless solution)	Colourless	Colourless	Pink	8.2 – 10.0
Methyl Orange	Red	Orange	Yellow	3.1 – 4.4
Turmeric (Natural Indicator)	Yellowish brown	No change (Yellowish brown)	Reddish brown	—
Universal Indicator / pH Paper	Red / Orange (0 – 6)	Green (7)	Blue / Violet (8 – 14)	0 – 14

K1A. LITMUS TEST

➤ Using blue and red litmus paper.

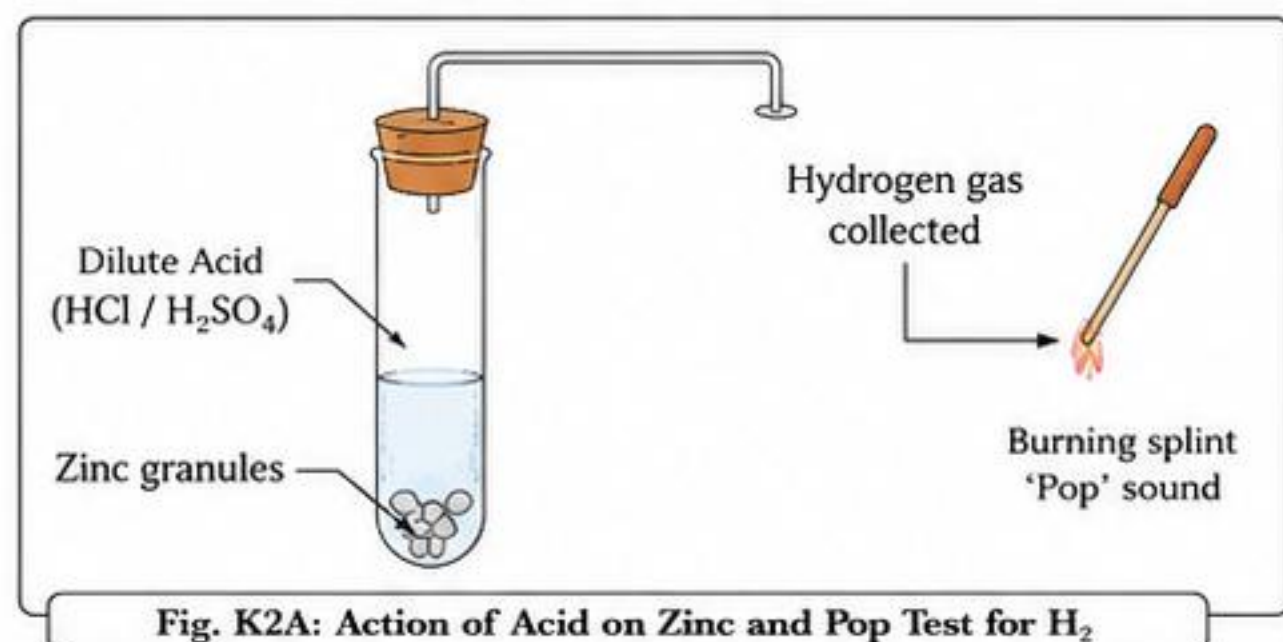


- If blue litmus turns red → solution is acidic.
- If red litmus turns blue → solution is basic.
- If no change in either → solution is neutral.

K2. TESTS BASED ON CHEMICAL REACTIONS

(a) Action on Metals (with Zinc)

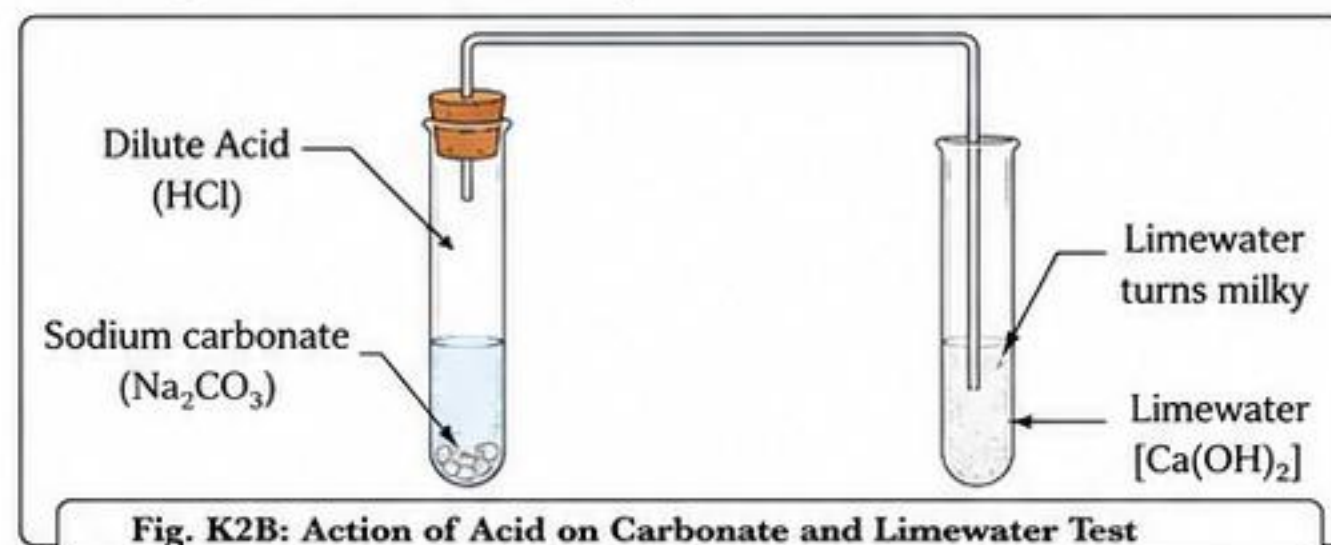
- Acids react with active metals like zinc to produce a salt and hydrogen gas.
- Bases do not react with zinc.
- This test helps identify acidic solutions.



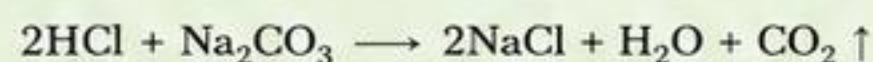
- **Observation:**
Effervescence (bubbles) is seen and a 'pop' sound is heard when a burning splint is brought near the gas.

(b) Action on Carbonates / Bicarbonates

- Acids react with carbonates or bicarbonates to produce salt, water and carbon dioxide gas.
- This gas turns limewater milky.



- **Observation:**
Brisk effervescence and limewater turns milky due to CO₂.
- **Equation:**



K3. USING pH PAPER / UNIVERSAL INDICATOR PAPER

- Dip pH paper in the solution and match the colour with the pH colour chart.
- Gives approximate pH value.
- pH < 7 → Acidic, pH = 7 → Neutral, pH > 7 → Basic.

Universal Indicator Colour Chart (Reference)

pH	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Colour	Red	Orange	Yellow	Yellow-Green	Green	Blue	Violet								

K4. OBSERVATION-BASED IDENTIFICATION SUMMARY

Test Performed	Acidic Solution	Basic Solution	Neutral Solution
Blue Litmus	Turns Red	No change (Blue)	No change (Blue)
Red Litmus	No change (Red)	Turns Blue	No change (Red)
Phenolphthalein	Colourless	Pink	Colourless
Methyl Orange	Red	Yellow	Orange
Turmeric	Yellowish brown	Reddish brown	No change
With Zinc	Effervescence, H ₂ gas evolved	No reaction	No reaction
With Na ₂ CO ₃	Effervescence, CO ₂ evolved; Limewater turns milky	No reaction	No reaction
pH Paper	pH < 7 (0–6)	pH > 7 (8–14)	pH = 7 (Green)

★ EXAM TIP BOX – HOW TO WRITE OBSERVATIONS & INFERENCE

- Write observations in the correct order of the test performed.
- Use standard terms: colour change, effervescence, gas evolved, no change, etc.
- Mention the indicator and its result clearly.
- State the inference in a clear sentence.
- If equations are asked, write the balanced equation.

Example Answer Format:

Test with blue litmus: Turns red.
 Test with zinc: Brisk effervescence; 'pop' sound with burning splint.
 Test with sodium carbonate: Brisk effervescence; limewater turns milky.
Inference: The given solution is acidic.

CLASS 10 SCIENCE – CHAPTER 2

ACIDS, BASES AND SALTS

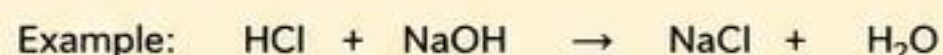
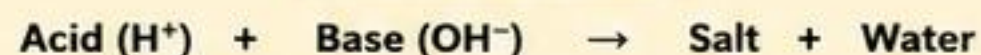
PART 2

SECTION L – NEUTRALISATION IN DAILY LIFE AND REAL-LIFE APPLICATIONS

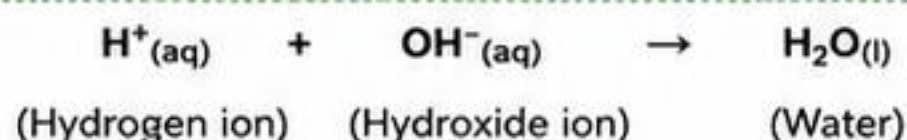
1. NEUTRALISATION – CONCEPT

- When an acid reacts with a base, they **neutralise** each other and form salt and water.
- This reaction is called **neutralisation**.
- It removes the acidic or basic nature.

GENERAL EQUATION



IONIC VIEW OF NEUTRALISATION



KEY POINTS

- ✓ Neutralisation is an exothermic reaction.
- ✓ pH of the resulting solution is around 7 (neutral).
- ✓ It forms salt and water.
- ✓ Used in many situations to remove excess acid or base.

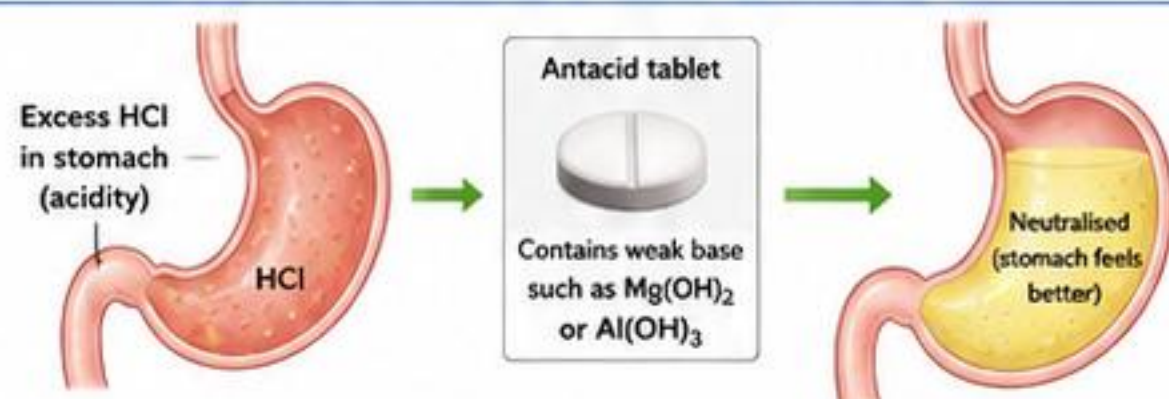
2. COMMON NEUTRALISATION EXAMPLES

- Hydrochloric acid + Sodium hydroxide $\rightarrow$ Sodium chloride + Water
- Sulphuric acid + Calcium hydroxide $\rightarrow$ Calcium sulphate + Water
- Nitric acid + Potassium hydroxide $\rightarrow$ Potassium nitrate + Water

NEUTRALISATION REACTION (MOLECULAR VIEW)

Acid	Base	Salt Formed	Water
HCl	NaOH	NaCl	H ₂ O
H ₂ SO ₄	Ca(OH) ₂	CaSO ₄	H ₂ O
HNO ₃	KOH	KNO ₃	H ₂ O
CH ₃ COOH	NaOH	CH ₃ COONa	H ₂ O

DIAGRAM 1: ANTACID ACTION IN STOMACH



REACTION (Example with Mg(OH)₂)



(Hydrochloric acid) (Magnesium hydroxide) (Magnesium chloride) (Water)

CENTUM FOCUS – EXAM POINTS (MUST WRITE)

- ★ Define neutralisation and write its general equation.
- ★ Write ionic equation: $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$.
- ★ Give at least 4–5 examples of neutralisation in daily life.
- ★ Draw neat labelled diagrams (Antacid action / Acidic soil / Sting relief / Tooth decay).
- ★ Write substances used in each case and their purpose.
- ★ Mention importance: health, agriculture, industry, environment, hygiene.



3. NEUTRALISATION IN DAILY LIFE – PROBLEMS, CAUSES & REMEDIES

Problem / Situation	Cause (Acidic / Basic)	Remedy (Neutralisation)	Example of Substance Used
Indigestion / Acidity	Excess HCl in stomach	Antacid neutralises excess acid	Mg(OH) ₂ , Al(OH) ₃ , NaHCO ₃
Acidic Soil	Excess acid due to heavy rainfall or fertilisers	Add basic substances to neutralise	Slaked lime (Ca(OH) ₂)
Factory Wastes	Acidic wastes released	Treated with bases before disposal	Ca(OH) ₂ , NaOH
Insect Stings (Bee / Ant)	Formic acid (mild acid)	Apply mild base to relieve pain	Baking soda (NaHCO ₃), Calamine
Self-Defence Chemicals	Acidic / basic chemicals used by attackers	Neutralising agents used to reduce effect	Baking soda, Vinegar, Lemon juice
Tooth Decay	Acid produced by bacteria	Basic toothpaste neutralises acid	Calcium carbonate (CaCO ₃)
Water Treatment	Acidic / basic water	Neutralised to safe pH (6.5 – 8.5)	Lime, NaOH, Alum
Environment	Acid rain / acidic emissions	Neutralised to protect life and soil	Lime (CaCO ₃), Ca(OH) ₂

DIAGRAM 2: ACIDIC SOIL TREATED WITH SLAKED LIME

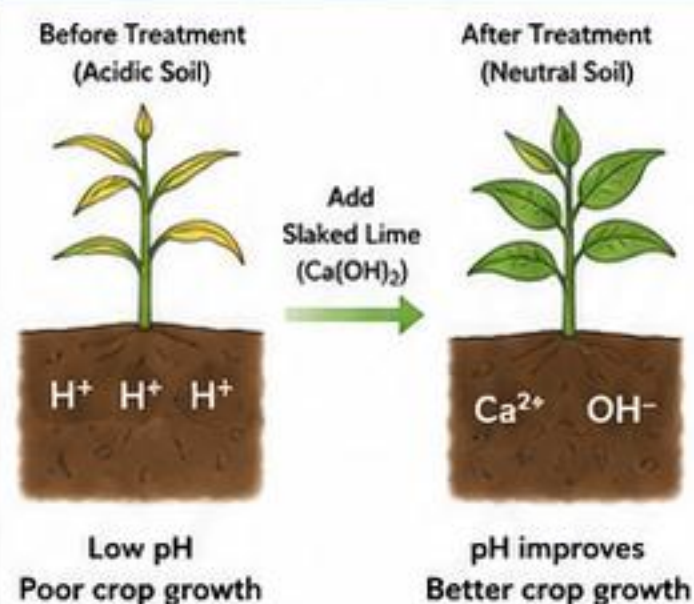
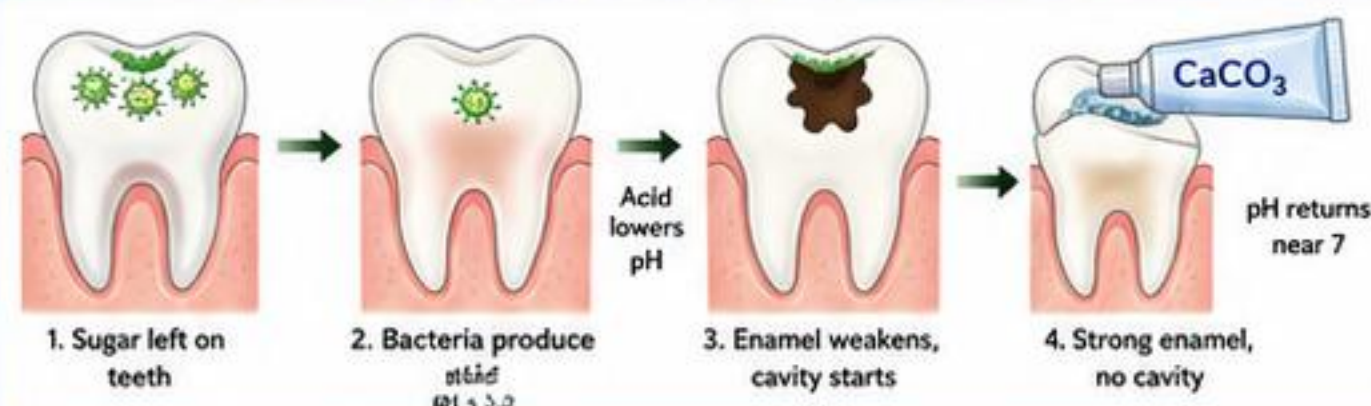


DIAGRAM 3: INSECT STING AND RELIEF



DIAGRAM 4: TOOTH DECAY AND NEUTRALISATION



ACID / BASE USED IN EACH CASE

Situation	Acid Involved	Base Used for Neutralisation
Indigestion	Hydrochloric acid (HCl)	Mg(OH) ₂ / Al(OH) ₃ / NaHCO ₃
Acidic Soil	Acidic components (H ⁺)	Ca(OH) ₂ (Slaked lime)
Insect Sting	Formic acid (HCOOH)	NaHCO ₃ / Calamine (basic)
Tooth Decay	Lactic acid	CaCO ₃ (in toothpaste)
Water Treatment	Excess acids / bases	Lime, NaOH, Alum etc.

CHAPTER 2: ACIDS, BASES AND SALTS

Part 2 of 2 – Complete Study Notes

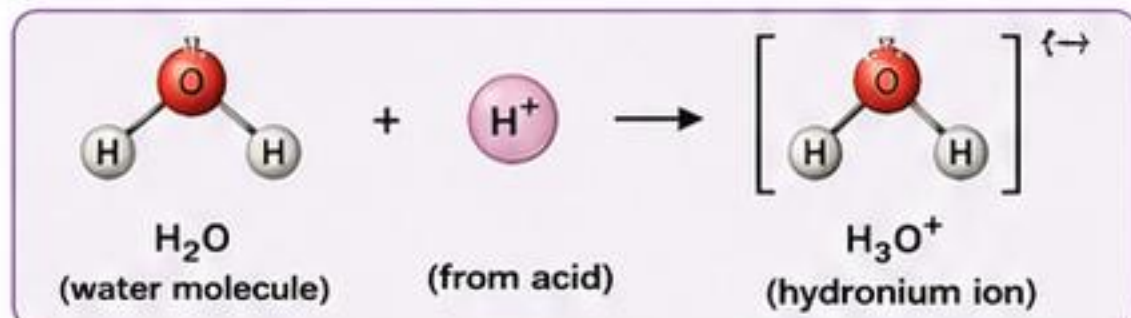
SECTION M – EXTRA DEPTH: IONISATION, HYDRONIUM, HYDROXIDE AND WHY WATER IS ESSENTIAL

1. IONISATION OF ACIDS AND BASES IN WATER

- Acids and bases produce ions only when they dissolve in water.
- Water molecules surround (solvate) the particles and help to break bonds, releasing ions.
- These ions are responsible for acidity, basicity, conductivity and indicator change.

2. HYDRONIUM ION FORMATION (H_3O^+)

- When an acid dissolves in water, the H^+ ion does not exist free. It immediately combines with a water molecule.
- $\text{H}^+ + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+$ (Hydronium ion)
- H_3O^+ is the actual acidic species in aqueous solutions.



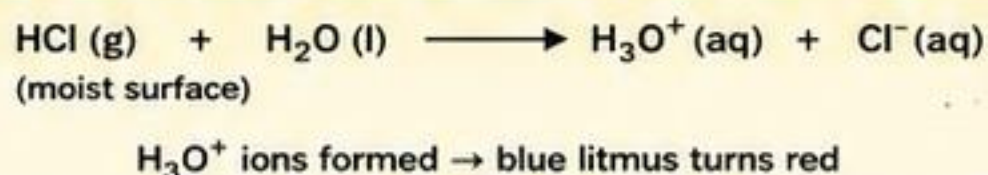
3. HYDROXIDE ION FORMATION (OH^-)

- Bases in water release hydroxide ions.
- OH^- ions are responsible for basic properties.
- They increase the concentration of OH^- in solution.



4. WHY DRY HCl DOES NOT AFFECT DRY BLUE LITMUS?

- In dry state, HCl gas does not ionise; it has no H_3O^+ ions.
- Without ions, it cannot show acidic character.
- When moist, HCl dissolves in water present on litmus paper:



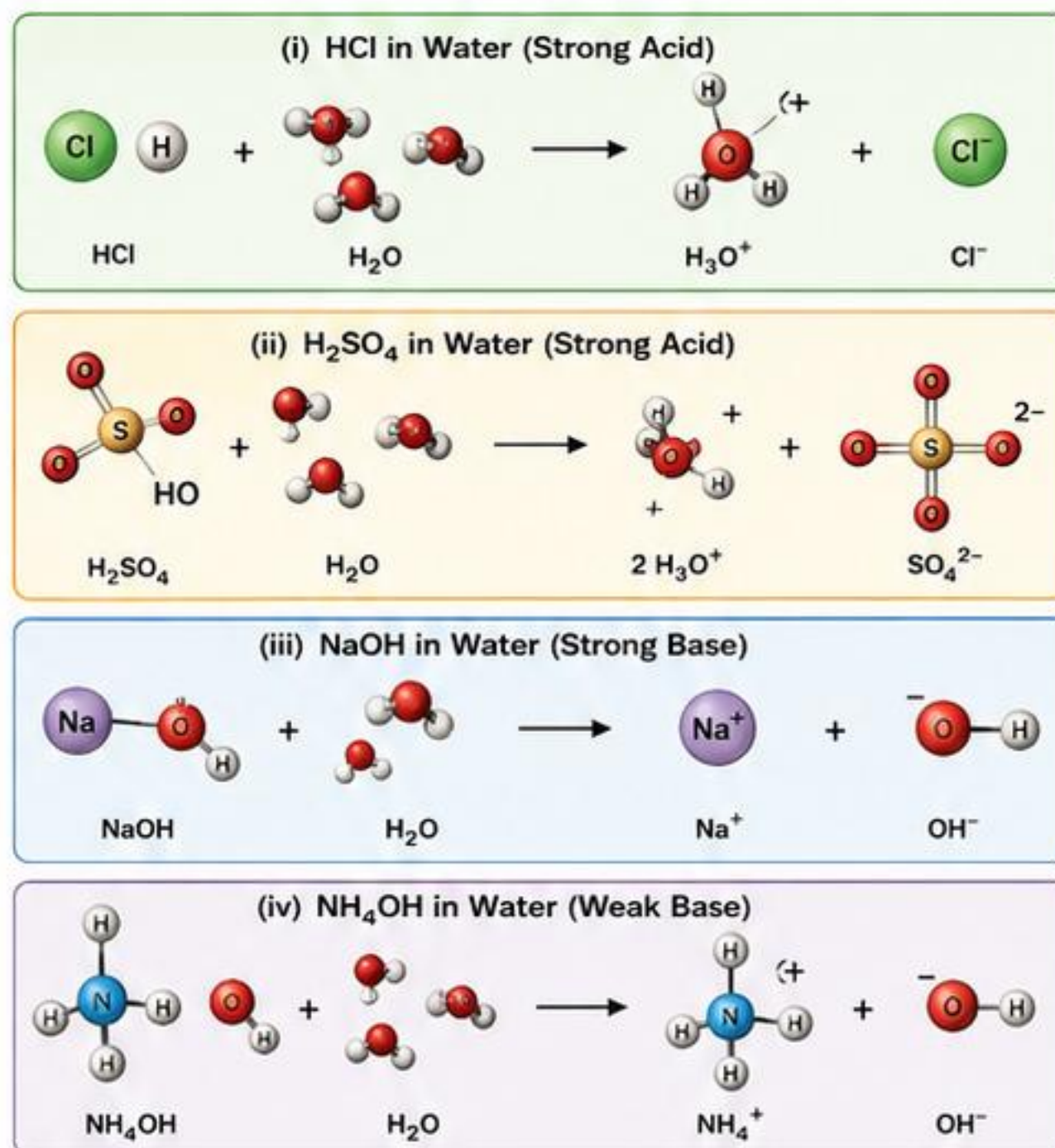
5. WHY SUBSTANCES SHOW ACIDIC/BASIC PROPERTIES ONLY IN AQUEOUS SOLUTION?

- Ions are formed only in water (due to ionisation).
- In solid or gas state, ions are absent $\rightarrow$ no acidic/basic behaviour.
- Water is essential because:
 - It separates ions (ion-dipole attraction).
 - It stabilizes ions (solvation).
 - It allows ions to move freely.
 - It enables reactions and indicator changes.

Key Takeaway

Acids and bases need water to ionise.
Water provides ions, mobility and reactions—
without water, no acidic or basic properties.

6. IONISATION OF COMMON ACIDS AND BASES IN WATER



7. ELECTROLYTES AND CONDUCTIVITY IN ACIDIC/BASIC SOLUTIONS

- Acids, bases and salts in water produce ions $\rightarrow$ electrolytes.
- Electrolytes conduct electricity in aqueous or molten state.

Type of Electrolyte	Ionisation	Conductivity	Example
Strong electrolytes	Complete	High (bright bulb)	HCl, NaOH, H_2SO_4
Weak electrolytes	Partial	Low (dim bulb)	CH_3COOH , NH_4OH
Non-electrolytes	No ionisation	No conduction	Sugar, alcohol, urea

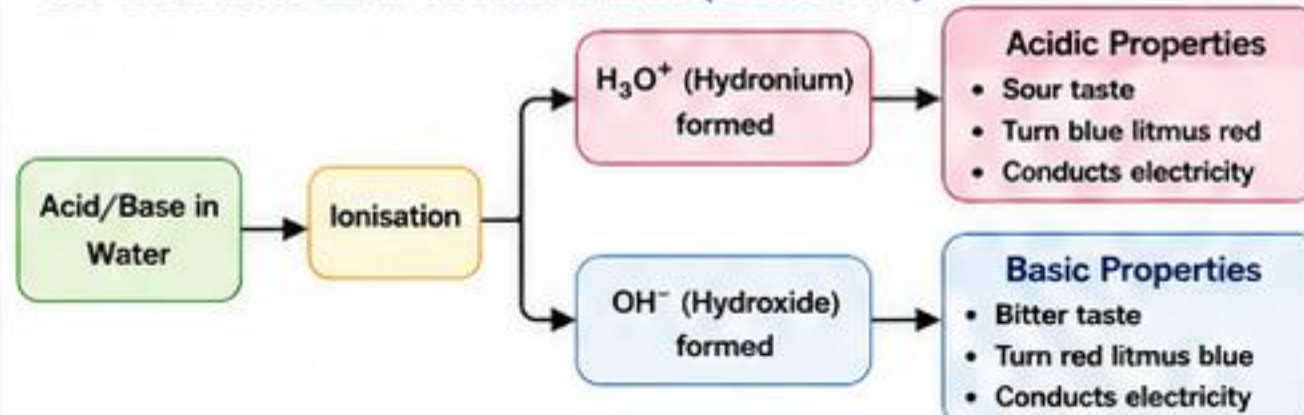
8. RELATION BETWEEN IONS AND INDICATOR CHANGE

Ions present in solution	Solution nature	Effect on indicators
H_3O^+ (Hydronium) $>$ OH^-	Acidic	Blue litmus $\rightarrow$ Red Methyl orange $\rightarrow$ Red Phenolphthalein $\rightarrow$ Colourless
OH^- (Hydroxide) $>$ H_3O^+	Basic	Red litmus $\rightarrow$ Blue Methyl orange $\rightarrow$ Yellow Phenolphthalein $\rightarrow$ Pink
$\text{H}_4\text{O}^+ \approx \text{OH}^-$	Neutral	No change in indicators

9. COMPARISON OF ACIDS AND BASES

Property	Strong Acid	Weak Acid	Strong Base	Weak Base
Ionisation in water	Complete	Partial	Complete	Partial
Ions produced	High $[\text{H}_3\text{O}^+]$	Low $[\text{H}_3\text{O}^+]$	High $[\text{OH}^-]$	Low $[\text{OH}^-]$
pH of solution	Very low (close to 0)	Low (above 0)	Very high (close to 14)	High (below 14)
Conductivity	High	Low	High	Low
Examples	HCl, H_2SO_4 , HNO_3	CH_3COOH , H_2CO_3	NaOH, KOH, $\text{Ca}(\text{OH})_2$	NH_4OH , $\text{Mg}(\text{OH})_2$

10. HOW IONS LEAD TO PROPERTIES (FLOWCHART)



HIGH-SCORE EXAM TIPS

- Always write ions, not atoms (H_3O^+ , OH^- , not H^+).
- Mention water in equations and explanations.
- Learn full names: Hydronium ion (H_3O^+), Hydroxide ion (OH^-).
- For dry HCl and litmus question: mention "no ionisation in dry state".
- For indicator questions, relate change to ions present.
- Use flowchart/schematic diagrams in answers.
- Write balanced ionic equations wherever possible.

CLASS 10 SCIENCE – CHAPTER 2

ACIDS, BASES AND SALTS (Part 2)

SECTION Q - COMPLETE FORMULA, EQUATION AND OBSERVATION MASTER SHEET

No.	Reaction / Concept	Balanced Equation (With States)	Reactants (Name)	Products (Name)	Type of Concept	Observation / Inference	Use / Importance
1.	Acid + Base (Neutralisation)	$\text{HCl (aq)} + \text{NaOH (aq)} \rightarrow \text{NaCl (aq)} + \text{H}_2\text{O (l)}$	Hydrochloric acid + Sodium hydroxide	Sodium chloride + Water	Neutralisation (Strong acid + Strong base)	No colour change (litmus: red $\rightarrow$ blue then violet). Heat may be felt (exothermic).	Antacid tablets neutralise excess stomach acid. Used in treating acidic soil, bee sting.
2.	Acid + Metal (Above H in activity series)	$\text{Zn (s)} + 2\text{HCl (aq)} \rightarrow \text{ZnCl}_2 \text{ (aq)} + \text{H}_2 \text{ (g)} \uparrow$	Zinc + Hydrochloric acid	Zinc chloride + Hydrogen gas	Displacement (Single replacement)	Brisk effervescence. Colourless, odourless gas evolved which burns with 'pop' sound.	Preparation of hydrogen gas in the laboratory (and industries).
3.	Acid + Metal (Dilute Acid)	$\text{Mg (s)} + \text{H}_2\text{SO}_4 \text{ (dil.)} \rightarrow \text{MgSO}_4 \text{ (aq)} + \text{H}_2 \text{ (g)} \uparrow$	Magnesium + Dilute sulphuric acid	Magnesium sulphate + Hydrogen gas	Displacement	Effervescence. Metal dissolves. Hydrogen gas evolved.	Demonstrates that metals above H liberate hydrogen from acids.
4.	Acid + Metal (Ionic form)	$2\text{H}^+ \text{ (aq)} + \text{Mg (s)} \rightarrow \text{Mg}^{2+} \text{ (aq)} + \text{H}_2 \text{ (g)} \uparrow$	Hydrogen ions + Magnesium	Magnesium ions + Hydrogen gas	Ionic Equation (Displacement)	Effervescence. Hydrogen gas $\uparrow$	Shows H^+ is responsible for reaction with reactive metals.
5.	Acid + Carbonate	$\text{CaCO}_3 \text{ (s)} + 2\text{HCl (aq)} \rightarrow \text{CaCl}_2 \text{ (aq)} + \text{H}_2\text{O (l)} + \text{CO}_2 \text{ (g)} \uparrow$	Calcium carbonate + Hydrochloric acid	Calcium chloride + Water + Carbon dioxide	Double Displacement (with gas formation)	Brisk effervescence. CO_2 turns limewater milky.	Removing scales (CaCO_3) with acids; preparation of CO_2 gas.
6.	Acid + Hydrogen carbonate (Bicarbonate)	$\text{NaHCO}_3 \text{ (s)} + \text{HCl (aq)} \rightarrow \text{NaCl (aq)} + \text{H}_2\text{O (l)} + \text{CO}_2 \text{ (g)} \uparrow$	Sodium hydrogen carbonate + Hydrochloric acid	Sodium chloride + Water + Carbon dioxide	Double Displacement (with gas formation)	Effervescence. CO_2 turns limewater milky.	Antacids: NaHCO_3 neutralises excess gastric acid and releases CO_2 .
7.	Metal Oxide + Acid	$\text{CuO (s)} + 2\text{HCl (aq)} \rightarrow \text{CuCl}_2 \text{ (aq)} + \text{H}_2\text{O (l)}$	Copper(II) oxide + Hydrochloric acid	Copper(II) chloride + Water	Neutralisation (oxide + acid)	Black solid (CuO) dissolves to give blue solution.	Basic oxides are neutralised by acids (e.g., treating acidity of soil).
8.	Non-metal Oxide + Base	$\text{CO}_2 \text{ (g)} + 2\text{NaOH (aq)} \rightarrow \text{Na}_2\text{CO}_3 \text{ (aq)} + \text{H}_2\text{O (l)}$	Carbon dioxide + Sodium hydroxide	Sodium carbonate + Water	Combination (Non-metal oxide + base)	No visible change. Solution remains colourless.	Absorption of CO_2 in soda-lime (NaOH) solutions; carbonation of soft drinks.
9.	Chlor-Alkali Process (Overall Reaction)	$2\text{NaCl (aq)} + 2\text{H}_2\text{O (l)} \rightarrow 2\text{NaOH (aq)} + \text{Cl}_2 \text{ (g)} \uparrow + \text{H}_2 \text{ (g)} \uparrow$	Sodium chloride + Water	Sodium hydroxide + Chlorine gas + Hydrogen gas	Electrolytic Decomposition	Greenish-yellow Cl_2 gas at anode; colourless H_2 gas at cathode; NaOH in solution.	Industrial method to prepare NaOH , Cl_2 , H_2 . Basis of many industries.
10.	Bleaching Powder Formation	$\text{Ca(OH)}_2 \text{ (s)} + \text{Cl}_2 \text{ (g)} \rightarrow \text{CaOCl}_2 \text{ (s)} + \text{H}_2\text{O (l)}$	Slaked lime + Chlorine gas	Bleaching powder + Water	Combination	White powder formed (bleaching powder). Chlorine smell.	Disinfectant for drinking water; bleaching agent in laundry.
11.	Sodium Hydrogen carbonate (Heating)	$2\text{NaHCO}_3 \text{ (s)} \xrightarrow{\Delta} \text{Na}_2\text{CO}_3 \text{ (aq)} + \text{H}_2\text{O (g)} + \text{CO}_2 \text{ (g)} \uparrow$	Sodium hydrogen carbonate	Sodium carbonate + Water vapour + Carbon dioxide	Thermal Decomposition	Effervescence at first; then residue remains. CO_2 gas $\uparrow$	Basis of baking soda rising effect in cakes and breads.
12.	Washing Soda Crystallisation	$\text{Na}_2\text{CO}_3 \text{ (aq)} + 10\text{H}_2\text{O (l)} \rightarrow \text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O (s)}$	Sodium carbonate solution + Water	Washing soda (Decahydrate)	Crystallisation	Colourless crystals of washing soda (soda crystals) separate.	Used in cleaning agents, water softening, and in industries.
13.	Plaster of Paris (POP) Formation	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O (s)} \xrightarrow{\Delta} \text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O (s)} + 1\frac{1}{2}\text{H}_2\text{O (g)}$	Gypsum (Plaster of Paris raw material)	Plaster of Paris + Water vapour	Thermal Dehydration	White powder formed on heating; water vapour lost.	Used in making moulds, toys, statues, false ceilings.
14.	POP Setting (Hardening)	$\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O (s)} + 1\frac{1}{2}\text{H}_2\text{O (l)} \rightarrow \text{CaSO}_4 \cdot 2\text{H}_2\text{O (s)}$	Plaster of Paris + Water	Gypsum (hard solid mass)	Combination (Hydration)	Heat evolved; mixture sets into a hard mass.	Used in casts for fractured bones, sculptures, etc.
15.	Copper Sulphate Hydration	$\text{CuSO}_4 \text{ (s)} + 5\text{H}_2\text{O (l)} \rightarrow \text{CuSO}_4 \cdot 5\text{H}_2\text{O (s)}$	Anhydrous copper sulphate + Water	Copper sulphate pentahydrate (blue crystals)	Combination (Hydration)	White powder turns blue. Heat evolved (dissolution exothermic).	Used in laboratories, agriculture (fungicide), and industrial applications.
16.	Copper Sulphate Dehydration	$\text{CuSO}_4 \cdot 5\text{H}_2\text{O (s)} \xrightarrow{\Delta} \text{CuSO}_4 \text{ (s)} + 5\text{H}_2\text{O (g)}$	Copper sulphate pentahydrate	Anhydrous copper sulphate + Water vapour	Thermal Decomposition (Dehydration)	Blue crystals turn white; water vapour evolves.	Drying agents; used in preparation of anhydrous salts.
17.	Ammonium Chloride Formation	$\text{NH}_3 \text{ (g)} + \text{HCl (g)} \rightarrow \text{NH}_4\text{Cl (s)} \downarrow$	Ammonia gas + Hydrogen chloride gas	Ammonium chloride (white solid)	Combination (Precipitation)	Dense white fumes form and deposit on cooler surfaces.	Industrially used in batteries, fertilizers, flux in soldering.

SYMBOL & STATE MEANINGS

- $\uparrow$ Gas evolved
- $\downarrow$ Precipitate formed
- (aq) Aqueous solution
- (s) Solid
- (l) Liquid
- (g) Gas
- Δ Heat
- $\rightarrow$ Yields / Produces

QUICK OBSERVATION GUIDE

- Effervescence (bubbles) = Gas evolved (CO_2 or H_2)
- Limewater turns milky = CO_2 gas present
- 'Pop' sound on bringing burning splint near gas = Hydrogen gas
- Blue colour with CuSO_4 = Water present ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$)
- White solid fumes = NH_4Cl formed
- Chlorine gas has greenish-yellow colour & pungent smell.

CONCEPT KEY

- NR – Neutralisation
- DR – Displacement Reaction
- CP – Combination Reaction
- DD – Double Displacement Reaction
- ED – Electrolytic Decomposition
- TD – Thermal Decomposition
- HY – Hydration
- DE – Dehydration

CLASS 10 SCIENCE – CHAPTER 2

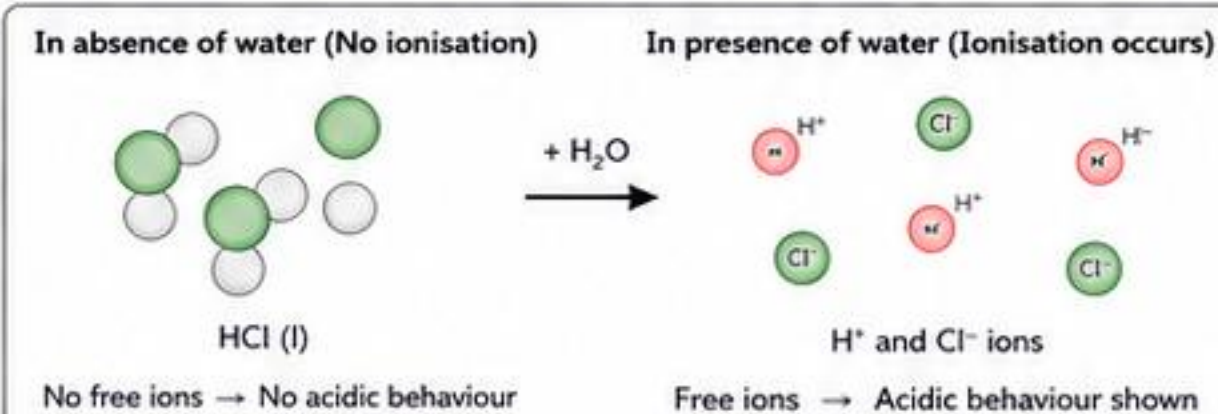
ACIDS, BASES AND SALTS

PART 2

SECTION 0 – NCERT BACK EXERCISE CONCEPT NOTES AND FREQUENTLY ASKED EXPLANATIONS

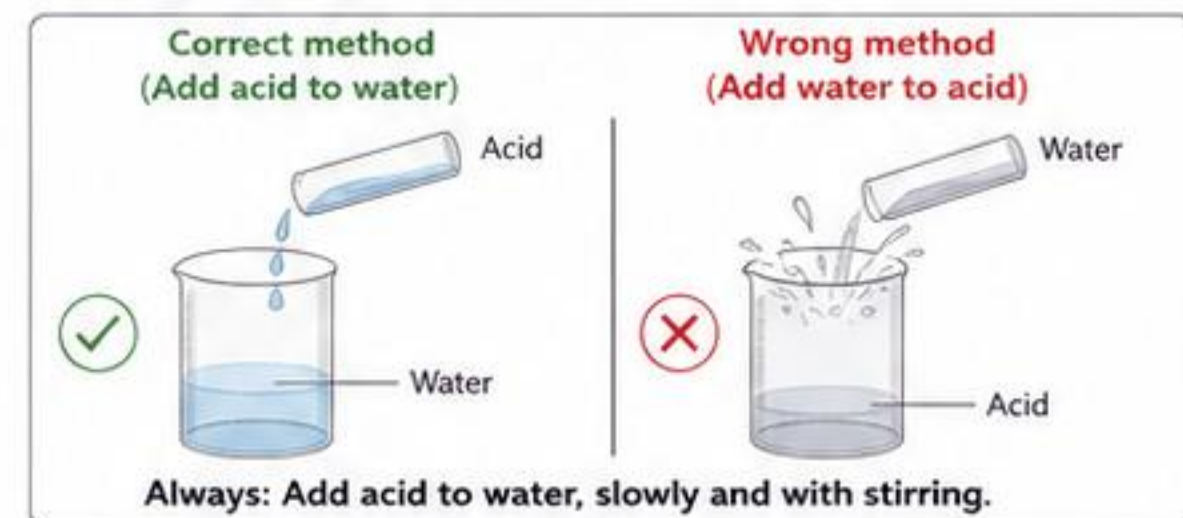
1. Why do acids not show acidic behaviour in the absence of water?

- Acids produce H^+ (or H_3O^+) ions in aqueous solution only.
- In absence of water, ions cannot form and there is no medium for ionisation.
- Therefore, acids do not show acidic behaviour without water.



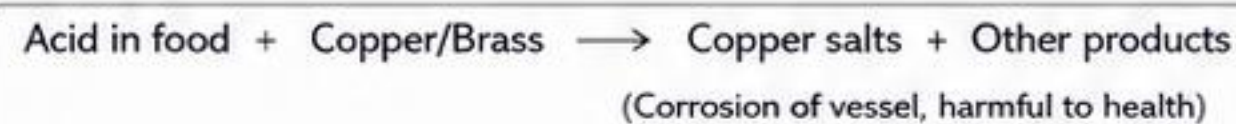
2. Why should acid be added to water and not water to acid while dilution?

- Dilution of acids is an exothermic process (heat is evolved).
- When acid is added to water, the large volume of water absorbs the heat and prevents splashing.
- If water is added to concentrated acid, heat evolved is at one spot and may cause sudden boiling and splashing of acid, leading to burns.



3. Why should curd or tamarind not be kept in brass or copper vessels?

- Curd and tamarind contain acids (lactic acid and tartaric/citric acid).
- These acids react with brass (an alloy of copper and zinc) or copper to form soluble salts.
- This corrodes the vessel and may make the food poisonous due to copper salts.
- Hence, such food items should not be stored in brass or copper vessels.



Use stainless steel, glass, or plastic for storing such food.

HOW TO WRITE 2-MARK AND 3-MARK ANSWERS

For 2-Mark Answers

- Write direct and to the point.
- Give the reason in 2–3 clear lines.
- Use key terms from the chapter.
- End with a concluding line.

Example (2-mark):

Acids do not show acidic behaviour in absence of water because they do not ionise without water. Without ions, there is no acidic nature.

For 3-Mark Answers

- Write in 3–4 well-structured points.
- Start with the reason.
- Add explanation/steps/examples.
- Conclude the answer.

Example (3-mark):

Acid should be added to water because dilution of acid is exothermic. When acid is added to a large volume of water, the heat is absorbed by water and no splashing occurs. If water is added to concentrated acid, heat is released at one spot causing violent boiling and splashing of acid.

4. Why does distilled water not conduct electricity, whereas acidic or basic solutions do?

- Distilled water contains almost no ions. It is made by boiling water and condensing the steam, which removes dissolved salts and impurities.
- Without ions, there are no charge carriers; hence it does not conduct electricity.
- Acidic or basic solutions contain ions (H^+ , OH^- , etc.) which conduct electricity.

Solution	Ionic particles present	Conducts electricity?
Distilled water	Almost none	No
Dilute HCl	H^+ , Cl^-	Yes
Dilute NaOH	Na^+ , OH^-	Yes

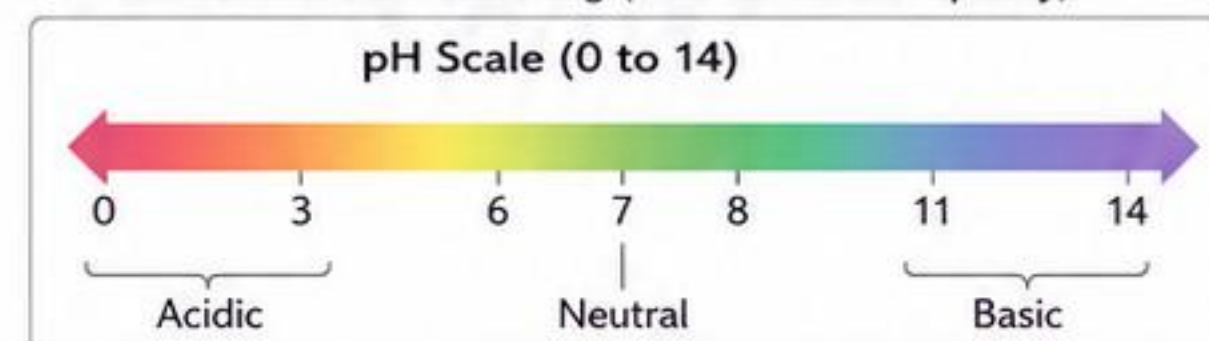
5. Why does an antacid tablet give relief from acidity?

- Acidity in the stomach is due to excess hydrochloric acid (HCl).
- Antacid tablets contain mild bases like $Mg(OH)_2$, $Al(OH)_3$, or $NaHCO_3$.
- These bases neutralise excess acid and raise the pH.
- This relieves pain and discomfort.



6. Why is pH important?

- pH tells how acidic or basic a solution is.
- It helps in:
 - ✓ Maintaining health (stomach pH, blood pH ~7.4).
 - ✓ Soil fertility (ideal pH for plant growth).
 - ✓ Industrial processes and quality control.
 - ✓ Environmental monitoring (acid rain, water quality).



7. Why is common salt (sodium chloride) important industrially?

- It is the raw material for the preparation of many chemicals:
 - ❖ Chlorine gas (used in bleaching, disinfectants, PVC)
 - ❖ Sodium hydroxide (NaOH) by chlor-alkali process
 - ❖ Hydrogen gas
 - ❖ Sodium carbonate and baking soda
- It is also used in food preservation, dyeing, textiles, and pharmaceutical industries.

8. Why are bleaching powder, washing soda and POP important?

Substance	Chemical Name / Formula	Uses (Importance)
Bleaching powder	Calcium oxychloride $CaOCl_2$	- Bleaching cotton and paper - Disinfecting drinking water - As oxidising agent
Washing soda	Sodium carbonate decahydrate $Na_2CO_3 \cdot 10H_2O$	- Used in cleaning soaps - Removes permanent hardness of water - Glass, paper, textile industry
POP	Plaster of Paris $CaSO_4 \cdot \frac{1}{2}H_2O$	- Used in setting fractures - Making casts, toys, decorative items, moulds

CLASS 10 SCIENCE – CHAPTER 2

ACIDS, BASES AND SALTS

PART 2

SECTION P - ASSERTION-REASON, CASE-BASED CONCEPTS AND COMMON TRAPS

◆ ASSERTION-REASON STYLE: KEY FACTS & TRAPS

In Assertion-Reason (A-R) questions, both statements may be true, but Reason may or may not explain Assertion.

Statement	Reason / Truth	Explanation
Acids turn blue litmus red and bases turn red litmus blue.	True	Basic property of litmus paper; due to change in H^+/OH^- concentration.
Stronger acids have higher concentration.	False	Strength depends on degree of ionisation, not concentration. (e.g., dilute HCl is stronger than dilute CH_3COOH).
pH of 7 is neutral.	True	At 25 °C, $[H^+] = [OH^-]$ so pH = 7 is neutral.
Pure water has pH 7 at all temperatures.	False	pH of pure water decreases below 7 when temperature increases.
Washing soda has water of crystallisation.	True	$Na_2CO_3 \cdot 10H_2O$ is washing soda.
Baking soda loses CO_2 when heated.	True	$2NaHCO_3 \rightarrow Na_2CO_3 + CO_2 + H_2O$
During neutralisation, salt and water are always formed.	True	Acid + Base $\rightarrow$ Salt + Water (in aqueous solutions).
Chlor-alkali process uses brine (NaCl solution) and electricity.	True	Electrolysis of brine produces Cl_2 , H_2 and NaOH.
Plaster of Paris sets because of water of crystallisation.	True	POP ($CaSO_4 \cdot \frac{1}{2}H_2O$) absorbs water and converts to gypsum ($CaSO_4 \cdot 2H_2O$).
All indicators show same colour in strong and weak acids.	False	Intensity of colour may differ; some indicators give different shades.
Dilute HCl + Zn gives H_2 gas.	True	Metal + Dilute Acid $\rightarrow$ Salt + H_2 (gas).
Dilute acid + metal carbonate gives CO_2 gas.	True	Acid + Carbonate $\rightarrow$ Salt + CO_2 + H_2O .



TIP: Look for the keyword in Reason – is it the correct scientific cause of Assertion? If not, the Reason does not explain the Assertion.

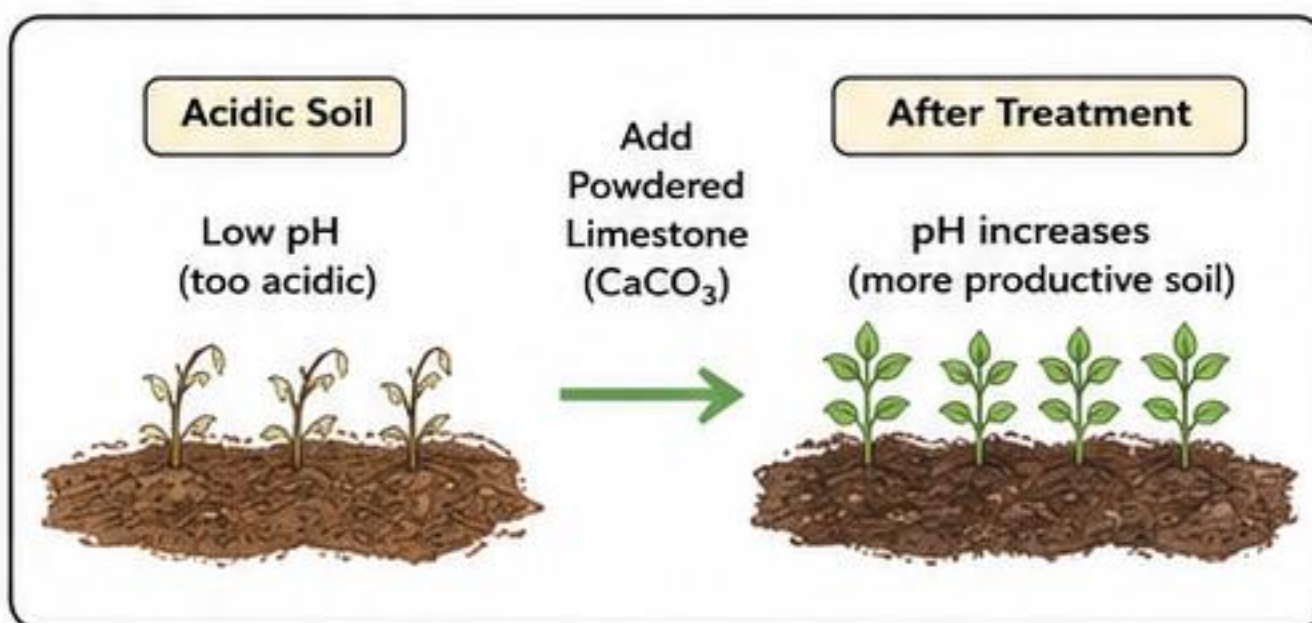


COMMON TRAPS STUDENTS MAKE

- Thinking strong acids have higher concentration.
- Assuming all bases are bitter and all acids are sour.
- Believing pH 7 means pure water at any temperature.
- Confusing indicators: e.g., phenolphthalein in acidic solution (should be colourless).
- Forgetting gases produced in reactions (CO_2 vs H_2).
- Thinking POP hardens by drying, not by chemical change.

◆ CASE-BASED CONCEPT 1: ACIDIC SOIL

Farmers noticed poor crop yield in a field. Soil test shows low pH (5.0). The soil is treated with powdered limestone.

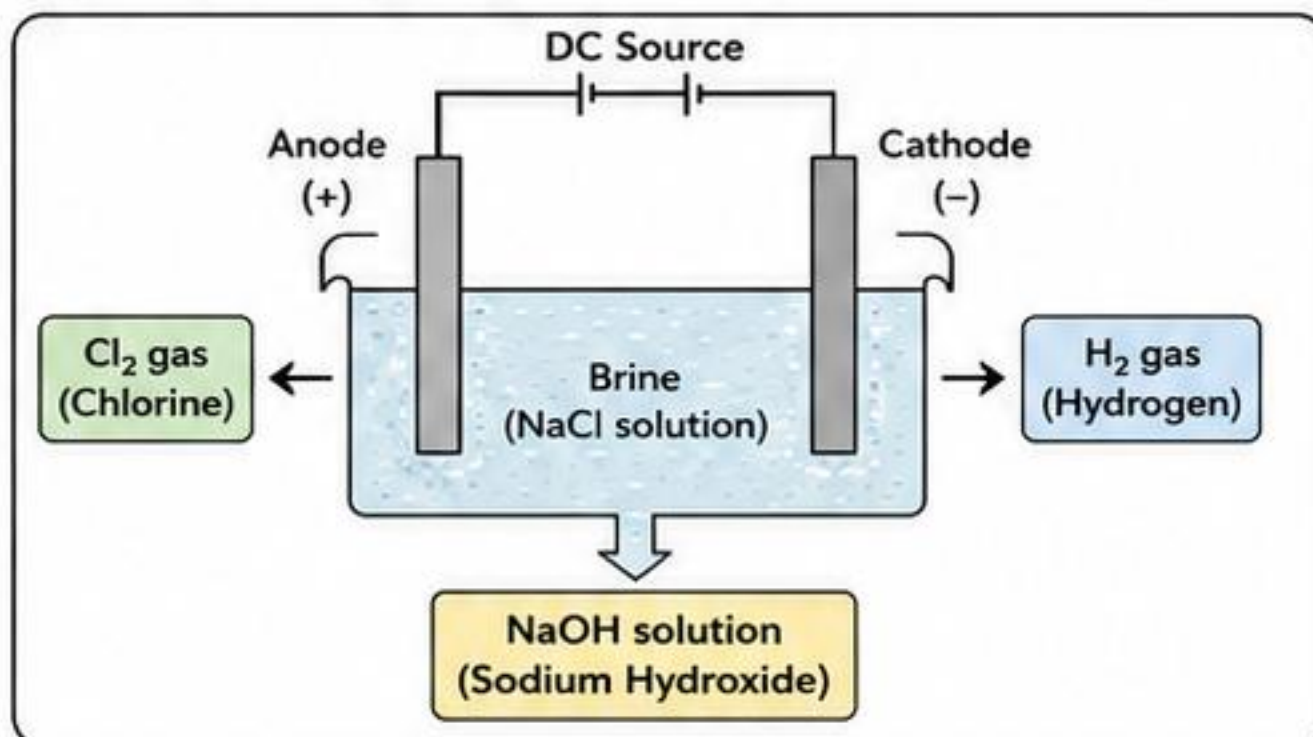


EXPLANATION NOTES:

- Acidic soil has excess H^+ ions.
- Limestone ($CaCO_3$) is basic in nature.
- It neutralises the acidity:
 $CaCO_3 + 2H^+ \rightarrow Ca^{2+} + CO_2 + H_2O$
- pH of soil increases and becomes more suitable for plant growth.
- This is called **liming** (adding lime/lime stone to soil).

◆ CASE-BASED CONCEPT 2: CHLOR-ALKALI PROCESS

Brine (concentrated NaCl solution) is electrolysed in a cell to produce useful products.



PRODUCTS:

- Chlorine gas (Cl_2) – used in bleaching, disinfecting water, PVC, etc.
- Hydrogen gas (H_2) – used in fuel, ammonia production.
- Sodium hydroxide (NaOH) – a strong base, used in soaps, detergents, paper, textiles.



CORRECT SCIENTIFIC WORDING

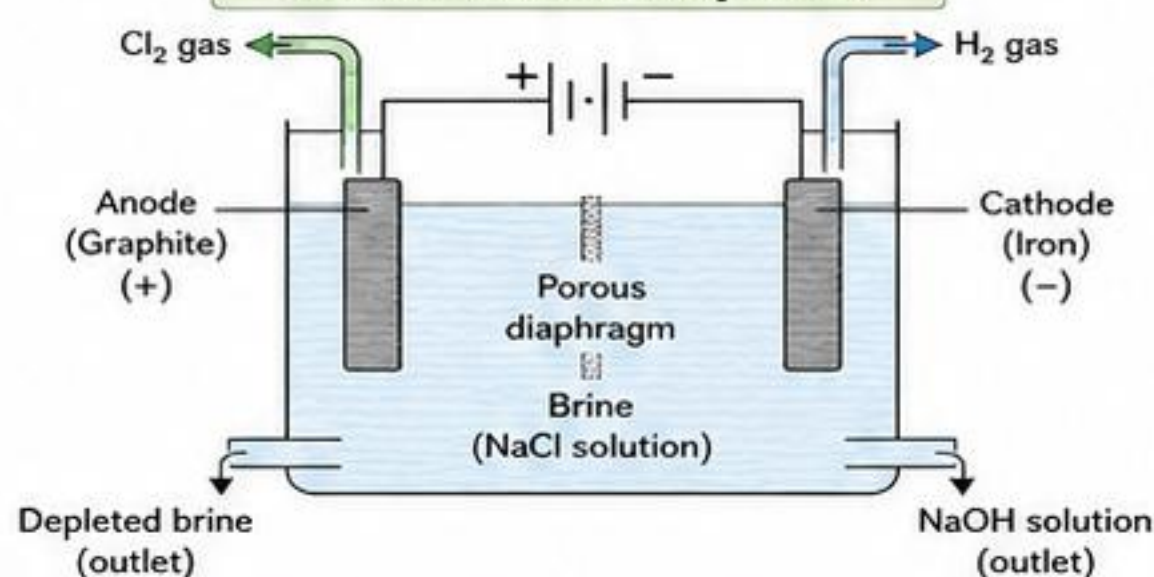
- An acid produces H^+ ions in aqueous solution.
- A base produces OH^- ions in aqueous solution.
- Strength of acid/base depends on degree of ionisation.
- Neutralisation is the reaction of an acid with a base to form salt and water.
- Water of crystallisation is water molecules chemically bound in the crystal.

CLASS 10 SCIENCE – STUDY NOTES

CHAPTER 2: ACIDS, BASES AND SALTS (PART 2)

SECTION R - DIAGRAM PRACTICE PAGE AND HOW TO DRAW IMPORTANT FIGURES

1. Chlor-Alkali Electrolytic Cell

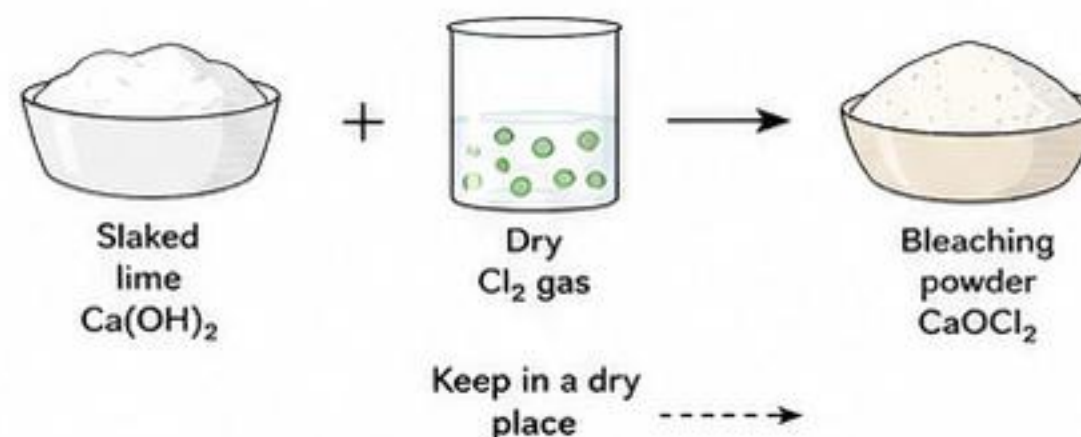


Label: Anode (+), Cathode (-), Brine, Porous diaphragm, Cl_2 gas, H_2 gas, NaOH solution, Depleted brine, Power supply.

Observation: Bubbles of Cl_2 at anode and H_2 at cathode. NaOH solution collects at cathode outlet.

Inference/Write: Electrolysis of brine produces Cl_2 , H_2 and NaOH.

2. Bleaching Powder – Preparation

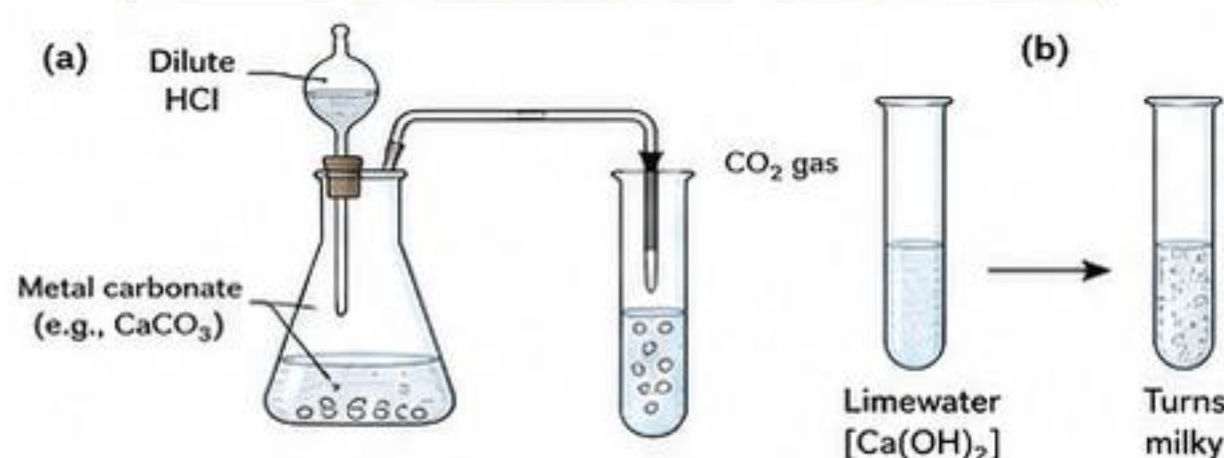


Label: Slaked lime Ca(OH)_2 , Dry Cl_2 gas, Bleaching powder CaOCl_2 , Keep in a dry place.

Observation: White powder of bleaching powder is obtained.

Inference/Write: Dry Cl_2 reacts with slaked lime to form bleaching powder (CaOCl_2), which should be stored in a dry place.

3. (a) Reaction of Metal Carbonate with Acid and (b) Limewater Test for CO_2



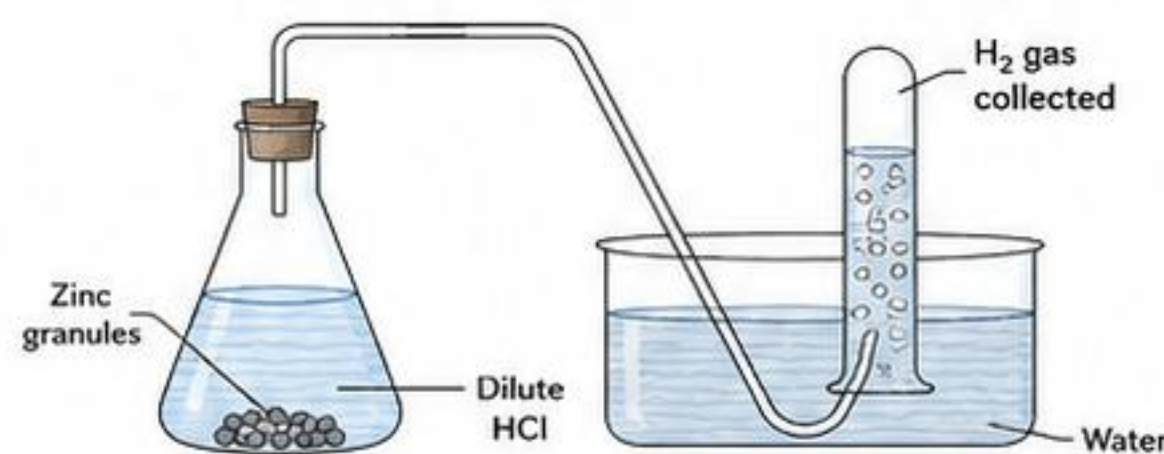
Label (a): Dilute HCl, Metal carbonate (e.g., CaCO_3), Flask, Gas delivery tube, CO_2 gas.

Label (b): Limewater [Ca(OH)_2], Turns milky.

Observation: (a) Effervescence of CO_2 is seen. (b) Limewater turns milky.

Inference/Write: Metal carbonate + acid $\rightarrow$ Salt + Water + CO_2 gas. CO_2 turns limewater milky due to formation of CaCO_3 .

4. Zinc with Dilute Acid – Hydrogen Collection



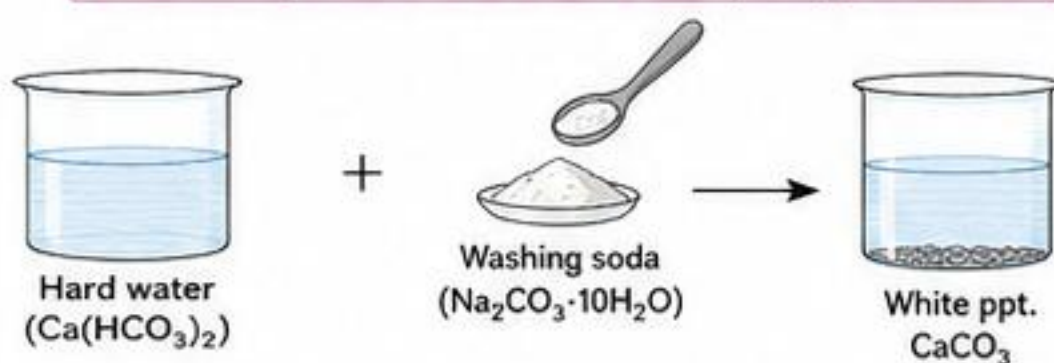
Label: Zinc granules, Dilute HCl, Cork, Delivery tube, Water trough, Inverted test tube, H_2 gas collected.

Observation: Effervescence is seen and colourless gas is collected.

Inference/Write: Zinc reacts with dilute acid to produce hydrogen gas.



5. Washing Soda – Removal of Temporary Hardness



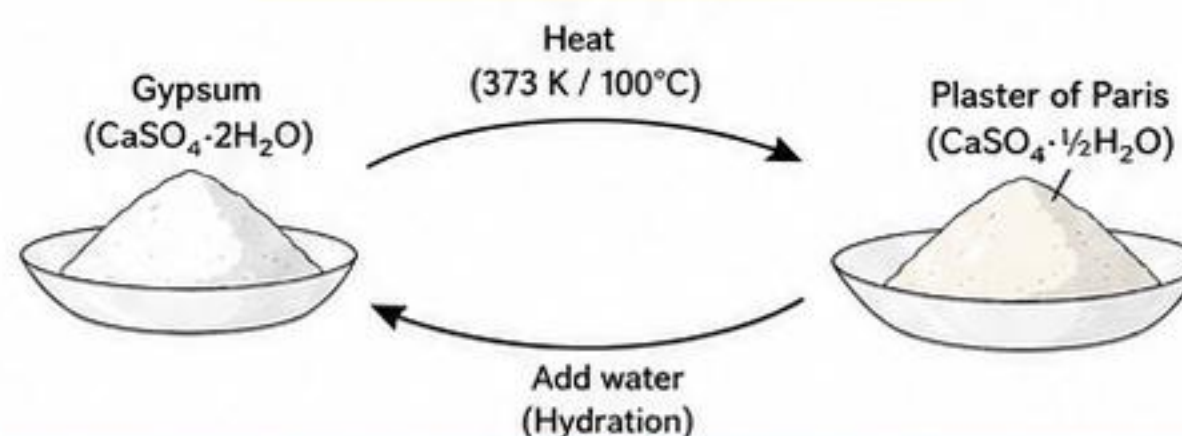
Label: Hard water ($\text{Ca(HCO}_3)_2$), Washing soda ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$), White ppt. CaCO_3 .

Observation: White precipitate forms.

Inference/Write: Washing soda removes temporary hardness by precipitating Ca^{2+} as CaCO_3 .



6. Plaster of Paris (POP) – Cycle

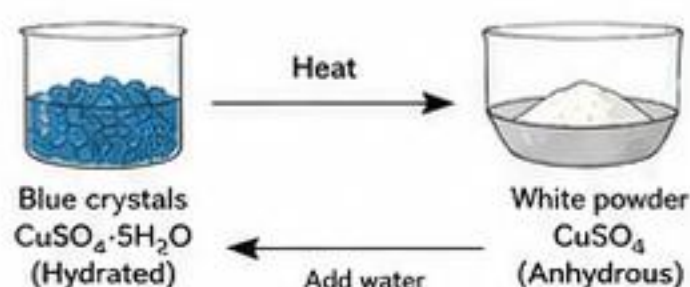


Label: Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$), Heat (373 K / 100°C), Plaster of Paris ($\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$), Add water (Hydration).

Observation: Gypsum on heating gives POP; POP on adding water sets to hard mass.

Inference/Write: Gypsum $\rightleftharpoons$ (heat) POP; POP + water $\rightarrow$ Hard solid.

7. Hydration and Dehydration of CuSO_4



Label: Blue crystals $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (Hydrated), Heat, White powder CuSO_4 (Anhydrous), Add water.

Observation: White presitals turn white on heating; white powder turns blue on adding water.

Inference/Write: $\text{CuSO}_4 \cdot 5\text{H}_2\text{O} \xrightarrow{\text{heat}} \text{CuSO}_4 + 5\text{H}_2\text{O}$; reverse on adding water.

8. Fire Extinguisher (CO_2 Type)



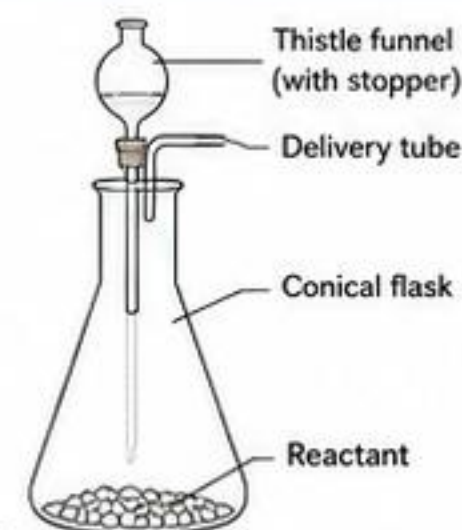
Label: Handle, Safety pin, Valve, Discharge tube, Nozzle, CO_2 cylinder.

Observation/Write: CO_2 is released as snow-like substance to put out fire.

Inference/Write: CO_2 does not support burning and smothers fire.

HOW TO DRAW SCIENCE DIAGRAMS NEATLY IN EXAMS

1. Use sharp pencil. Draw light lines first, then darken final lines.
2. Use ruler for straight lines and borders.
3. Make diagrams large and in the centre of the space provided.
4. Label with single horizontal lines. Do not cross lines.
5. Write labels clearly in block letters.
6. Show arrows to indicate gas flow, heat, movement, etc.
7. Underline the title of the diagram.
8. Keep diagrams simple, neat and proportionate.



CLASS 10 SCIENCE – CHAPTER 2

ACIDS, BASES AND SALTS

(PART 2)

SECTION S – ONE-MARK, TWO-MARK CONCEPT RECALL AND ULTRA-FAST REVISION TABLES

MUST-REMEMBER TERMS & DEFINITIONS

Term	One-Line Definition
Acid	A substance that releases H^+ ions in water; sour in taste.
Base	A substance that releases OH^- ions in water; bitter in taste and soapy to touch.
Salt	Product formed when an acid reacts with a base; usually neutral.
Neutralisation	Reaction between an acid and a base to form salt and water.
pH	Measure of acidity or basicity of a solution.
Indicator	A substance that shows different colours in acids and bases.
Litmus	Natural indicator (from lichens) used to test acids and bases.
Strong Acid	Completely ionises in water (e.g., HCl).
Weak Acid	Partially ionises in water (e.g., CH_3COOH).
Alkali	A base that is soluble in water and releases OH^- ions (e.g., NaOH).
Effervescence	Bubbles of a gas (CO_2) produced in a reaction.

IMPORTANT FORMULAS (Memorise!)

Concept	Formula / Representation
Neutralisation (general)	Acid + Base $\rightarrow$ Salt + Water
Hydrochloric acid	HCl
Sulphuric acid	H_2SO_4
Nitric acid	HNO_3
Sodium hydroxide (caustic soda)	NaOH
Calcium hydroxide (slaked lime)	$Ca(OH)_2$
Sodium chloride (common salt)	NaCl
Sodium hydrogen carbonate (baking soda)	$NaHCO_3$
Washing soda	$Na_2CO_3 \cdot 10H_2O$
Bleaching powder	$CaOCl_2$
Plaster of Paris	$CaSO_4 \cdot \frac{1}{2}H_2O$
Gypsum	$CaSO_4 \cdot 2H_2O$
POP setting	$CaSO_4 \cdot \frac{1}{2}H_2O + \frac{1}{2}H_2O \rightarrow CaSO_4 \cdot 2H_2O$

USES – QUICK RECALL

Substance	Uses
Hydrochloric acid (HCl)	Cleaning metals, leather industry, removing rust.
Sulphuric acid (H_2SO_4)	In car batteries, fertilizer industry, metal cleaning.
Sodium hydroxide (NaOH)	Making soap, paper, artificial fibres, drain cleaner.
Sodium hydrogen carbonate ($NaHCO_3$)	Baking (as baking soda), antacid, in fire extinguishers.
Washing soda (Na_2CO_3)	Washing clothes, cleaning hard water.
Bleaching powder ($CaOCl_2$)	Disinfecting water, bleaching cotton/linen.
Common salt (NaCl)	Food seasoning, preserving food.
Plaster of Paris ($CaSO_4 \cdot \frac{1}{2}H_2O$)	For setting fractures, making moulds, craft work.
Gypsum ($CaSO_4 \cdot 2H_2O$)	In wall boards, cement, blackboard chalk.

DAILY LIFE CONCEPT STRIP – INSTANT REVISION

Lemon  (Citric Acid) Sour; contains natural acid.	Vinegar  (Acetic Acid) Used in cooking; turns blue litmus red.	Soap  (Sodium/Other Salts of Fatty Acids) Made using base; cleans by removal of dirt and oil.	Baking Soda  ($NaHCO_3$) Antacid, baking, releases CO_2 with acids.	Table Salt  (NaCl) Neutral salt; used in food and preservation.	Plaster of Paris (POP)  ($CaSO_4 \cdot \frac{1}{2}H_2O$) Sets with water; used in casts and decorative items.	Washing Soda  ($Na_2CO_3 \cdot 10H_2O$) Softens hard water; used in washing clothes.	Bleaching Powder  ($CaOCl_2$) Disinfects water; bleaches clothes.	Gypsum  ($CaSO_4 \cdot 2H_2O$) Used in wall boards, cement, blackboard chalk.
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COLOUR CHANGES – INDICATORS IN ACIDS & BASES

Indicator	In Acidic Solution	In Neutral Solution	In Basic Solution
Litmus	Red	Purple	Blue
Red Litmus	Stays Red	Purple	Turns Blue
Blue Litmus	Turns Red	Purple	Stays Blue
Phenolphthalein	Colourless	Colourless	Pink
Methyl Orange	Red	Orange	Yellow
Turmeric (Natural)	Yellow	Yellow	Reddish Brown
China Rose Extract	Pink/Red	Purple	Green

pH SCALE – ULTRA-FAST RECALL

pH Value	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Nature	Acidic						Neutral		Basic (Alkaline)						

Key Points:

- pH < 7 $\rightarrow$ Acidic
- pH = 7 $\rightarrow$ Neutral (Pure water)
- pH > 7 $\rightarrow$ Basic / Alkaline
- Lower the pH $\rightarrow$ stronger the acid
- Higher the pH $\rightarrow$ stronger the base

DIFFERENCES – MUST MEMORISE

Feature	Acids	Bases
Taste	Sour	Bitter
Touch	Not slippery	Slippery/Soapy
Litmus	Blue litmus turns red	Red litmus turns blue
Phenolphthalein	Colourless	Pink
pH	Less than 7	Greater than 7
Examples	HCl, H_2SO_4 , CH_3COOH	NaOH, $Ca(OH)_2$, NH_4OH
Reaction with metals	Usually gives H_2 gas	Generally no reaction
Reaction with carbonates	Gives CO_2 gas	No CO_2 gas
In nature	Many are corrosive	Many are caustic
Feel	Burning sensation	Soapy/slippery feel

ONE-LINE EXPLANATIONS (EXAM BOOSTER)

- Acids turn blue litmus red because they release H^+ ions in water.
- Bases turn red litmus blue because they release OH^- ions in water.
- Neutralisation reactions are important to treat acidity in the stomach.
- Baking soda is used in fire extinguishers because it releases CO_2 .
- Plaster of Paris sets into a hard mass due to addition of water.
- Bleaching powder is kept in dry, airtight containers as it decomposes in presence of moisture and CO_2 .
- Hard water is softened by washing soda because it removes Ca^{2+} and Mg^{2+} ions.



SECTION T - FINAL HIGH-SCORE REVISION SHEET AND COMPLETE CHAPTER CONCEPT MAP



CLASS 10 SCIENCE – CHAPTER 2: ACIDS, BASES AND SALTS (PART 2 OF 2)

1. MOST IMPORTANT DEFINITIONS (with Examples)

- Acid:** A substance that turns blue litmus red in aqueous solution and produces $H^+(aq)$ ions. (Ex: HCl , CH_3COOH)
- Base:** A substance that turns red litmus blue in aqueous solution and produces $OH^-(aq)$ ions. (Ex: $NaOH$, $Ca(OH)_2$)
- Salt:** An ionic compound formed when an acid reacts with a base. (Ex: $NaCl$, $CuSO_4$)
- Neutralisation:** Reaction between an acid and a base to form salt and water. (Ex: $HCl + NaOH \rightarrow NaCl + H_2O$)
- Indicator:** A substance that shows different colours in acidic and basic solutions. (Ex: Litmus, Phenolphthalein)
- pH:** Measure of acidity/basicity of a solution (0–14 scale).
pH = 7 neutral; <7 acidic; >7 basic.
- Metal:** Substance that reacts with dilute acids to produce salt and hydrogen gas. (Ex: Zn , Mg)
- Displacement Reaction:** More reactive metal displaces less reactive metal from its salt solution.

2. TOP FORMULAS & EQUATIONS

A. Key Formulas

- $pH = -\log_{10} [H^+]$
- $pOH = -\log_{10} [OH^-]$
- $pH + pOH = 14$ (at $25^\circ C$)
- $Salt + Water \rightleftharpoons Acid/Base$ (Hydrolysis)

B. Important Equations

- Acid + Base $\rightarrow$ Salt + Water**
 $HCl + NaOH \rightarrow NaCl + H_2O$
- Metal + Dilute Acid $\rightarrow$ Salt + H_2**
 $Zn + 2HCl \rightarrow ZnCl_2 + H_2 \uparrow$
- Metal Oxide + Acid $\rightarrow$ Salt + Water**
 $CuO + 2HCl \rightarrow CuCl_2 + H_2O$
- Metal Hydroxide + Acid $\rightarrow$ Salt + Water**
 $NaOH + HCl \rightarrow NaCl + H_2O$
- Carbonate/Bicarbonate + Acid $\rightarrow$ Salt + Water + CO_2**
 $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2 \uparrow$
- Sulphite + Acid $\rightarrow$ Salt + Water + SO_2**
 $Na_2SO_3 + 2HCl \rightarrow 2NaCl + H_2O + SO_2 \uparrow$

3. COMPACT COMPARISONS

A. Acids vs Bases

Feature	Acids	Bases
Taste	Sour	Bitter
Touch	May irritate skin	Slippery
Litmus	Blue $\rightarrow$ Red	Red $\rightarrow$ Blue
pH	< 7	> 7
Examples	HCl , CH_3COOH	$NaOH$, $Ca(OH)_2$
Reaction with Metals	Produce H_2 gas	Generally no H_2 gas

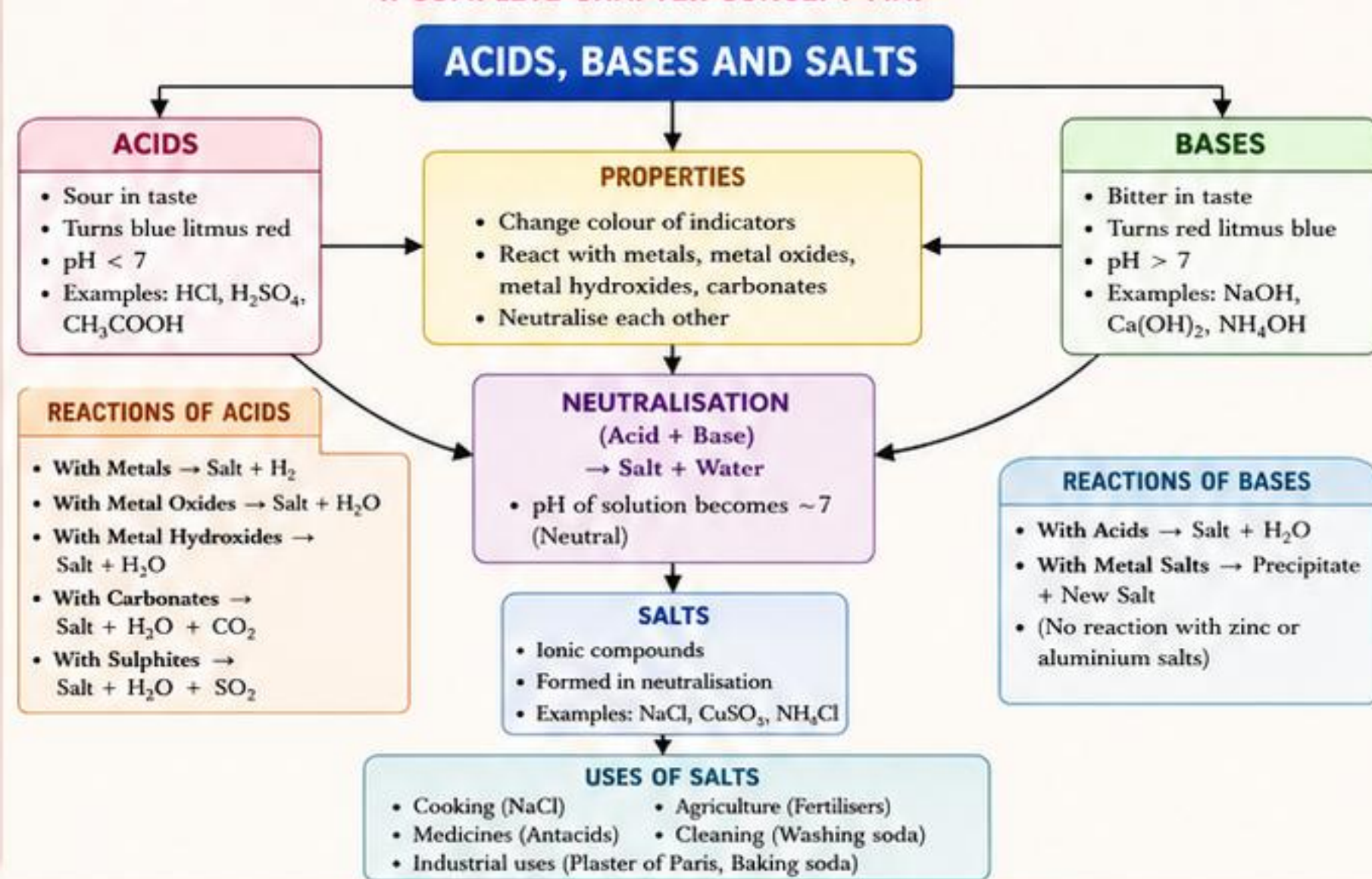
B. Strong vs Weak Acids/Bases

Type	Ionisation in Water	Examples
Strong Acids	Completely ionise	HCl , H_2SO_4
Weak Acids	Partially ionise	CH_3COOH , H_2CO_3
Strong Bases	Completely ionise	$NaOH$, KOH
Weak Bases	Partially ionise	NH_4OH

C. Types of Salts

Type of Salt	Formed From	Examples
Normal Salt	Strong acid + Strong base	$NaCl$
Acid Salt	Strong acid + Weak base	NH_4Cl
Basic Salt	Weak acid + Strong base	CH_3COONa
Double Salt	Two simple salts	Mohr's salt ($FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$)

4. COMPLETE CHAPTER CONCEPT MAP



5. TOP-SCORING KEYWORDS

- Acidity** – presence of $H^+(aq)$ ions in a solution.
- Basicity** – presence of $OH^-(aq)$ ions in a solution.
- Neutralisation** – acid + base $\rightarrow$ salt + water.
- Indicator** – shows colour change in acids/bases.
- pH** – measure of acidity/basicity (0–14).
- Salt** – product of neutralisation (ionic compound).
- Displacement** – more reactive metal displaces less reactive metal from its salt solution.
- Hydrolysis** – reaction of salt with water to form acid/base.

6. LAST-MINUTE CHECKLIST

- ☒ I can define acids, bases, salts and neutralisation.
- ☒ I can write chemical equations for all important reactions.
- ☒ I can identify acids/bases using indicators and pH.
- ☒ I can classify salts (normal, acid, basic, double).
- ☒ I can explain the uses of common salts.
- ☒ I can balance key equations correctly.
- ☒ I can relate pH values to acidity/basicity.
- ☒ I can answer long/short/Application questions confidently.

7. COMMON MISTAKES TO AVOID

- ☒ Thinking all bases turn phenolphthalein pink. (Only strong bases do.)
- ☒ Writing equation without balancing.
- ☒ Forgetting to write the state symbols (aq, s, g, l).
- ☒ Confusing pH and pOH.
- ☒ Believing bases react with zinc to give H_2 gas. (They don't.)
- ☒ Forgetting CO_2 or SO_2 in carbonate/sulphite reactions.
- ☒ Not mentioning indicator colour change clearly.

8. HOW TO WRITE HIGH-SCORING LONG ANSWERS (Very Short Guide)

- Start with a clear definition.
- Write relevant chemical equations.
- Explain the observation with reason.
- Give examples wherever possible.
- Conclude with the overall result/use.
- Use labelled diagrams if asked.
- Write neatly, in points/steps for clarity.

9. MINI-SUMMARY PANELS

INDICATORS & COLOURS			
Indicator	Acid	Base	Neutral
Litmus	Red	Blue	Violet
Phenolphthalein	Colourless	Pink	Colourless
Methyl Orange	Red	Yellow	Orange

pH SCALE (0 – 14)	IMPORTANT GASES
0 – 6 $\rightarrow$ Acidic 7 $\rightarrow$ Neutral 8 – 14 $\rightarrow$ Basic - Lower pH = more acidic - Higher pH = more basic	- From Acids + Metals $\rightarrow H_2$ - From Carbonates $\rightarrow CO_2$ - From Sulphites $\rightarrow SO_2$

10. CENTUM FOCUS – YOU CAN DO IT!



FOCUS

- Understand concepts, not just memorise. Practice equations daily. Revise indicators, pH and reactions again and again.

STRATEGY

- Solve NCERT examples, extra questions and previous year papers. Write full answers in practice tests.

ATTITUDE

- Stay calm, stay consistent, stay confident. Believe in yourself because hard work today brings success tomorrow!

THINK POSITIVE
WORK SMART
SCORE CENTUM! **100**